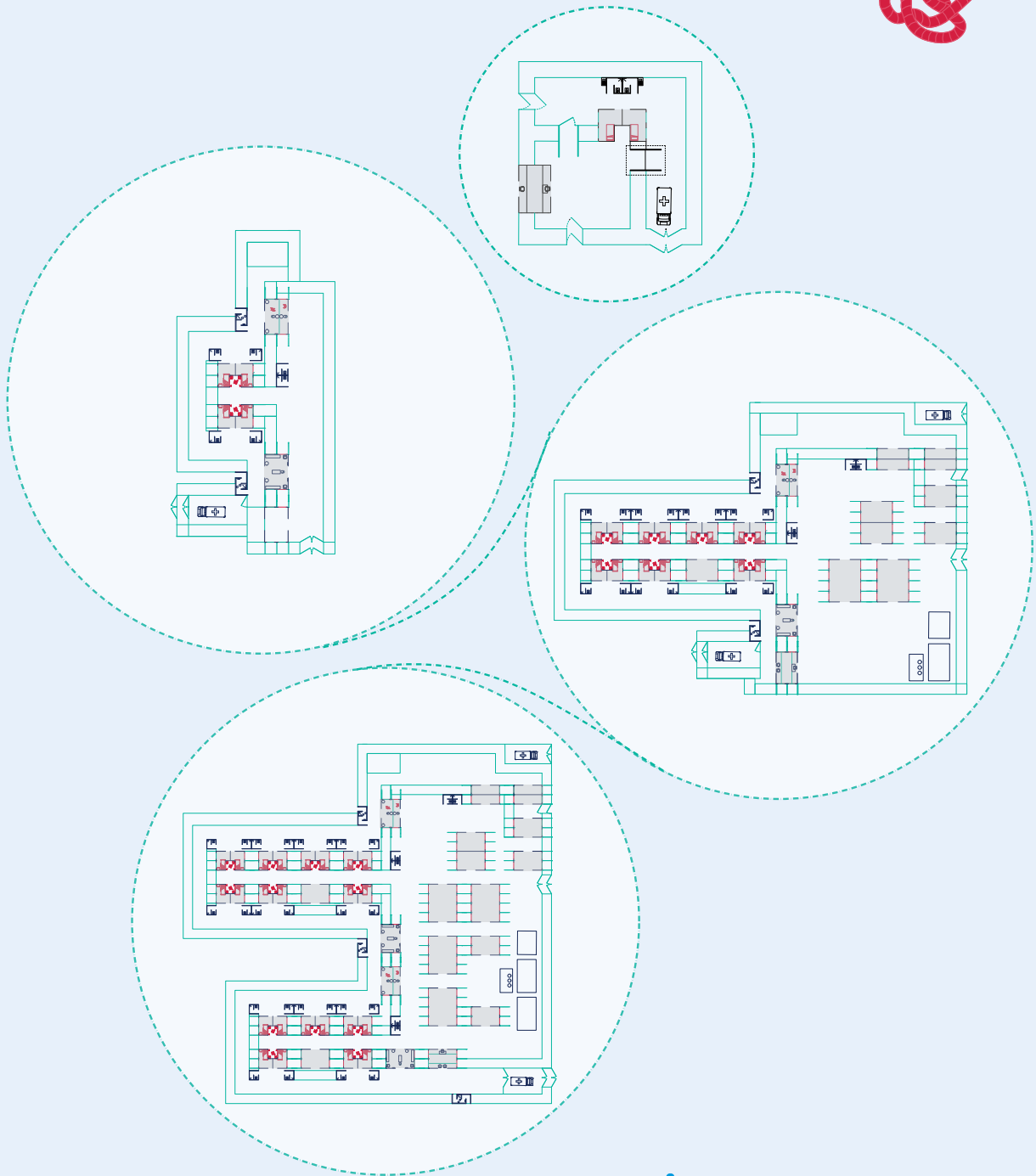
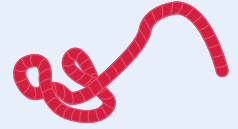


Technical specifications for filovirus disease treatment centres



World Health
Organization

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Abbreviations and acronyms

HCWs	Health and care workers
IPC	Infection prevention and control
PPE	Personal protective equipment
WASH	Water, sanitation and hygiene
WHO	World Health Organization

Glossary

The definitions provided in this glossary denote specific physical spaces and operational activities within Ebola and Marburg treatment centres. For clarity and consistency throughout this document, these terms are used exclusively within the present operational context and may therefore differ from their conventional clinical definitions and usage.

Ambulance bay	Area reserved for the arrival of the ambulance and the transfer of patients (1).
Care workers	Care workers provide direct, personal care services in the home, in health care and residential settings, while assisting with routine tasks of daily life, and while performing a variety of other tasks of a simple and routine nature (2). • Health-care assistants: Institution-based, personal-care workers who provide direct, personal care and assistance with activities of daily living to patients and residents in a variety of health-care settings, such as hospitals, clinics and residential nursing-care facilities. They generally implement established care plans and practices under the direct supervision of medical, nursing, or other health professionals or associate professionals. • Home-based personal care workers: Workers who provide routine personal care and assistance with activities of daily living to persons in need of such care due to effects of ageing, illness, injury or other physical or mental conditions, in private homes and other independent, residential settings (2).
Children and family spaces	Children and family spaces are specialized areas designed to support the unique needs of children and families of patients suspected or confirmed with Ebola or Marburg diseases.
Clinical assessment	Clinical assessment is the systematic evaluation of a patient's health status through history-taking, physical examination, observation, and review of relevant investigations, to identify clinical problems, determine severity, guide diagnosis, and inform appropriate care or referral.
Confirmed patients' area	It is a designated section within the treatment centre specifically allocated for the isolation and clinical management of individuals who meet the case definition for confirmed case (3).
Doffing area	A specifically designated space within the treatment centre where health and care workers and other staff members take off personal protective equipment (PPE) (4).
Donning area	A specifically designated space within the treatment centre where health and care workers and other staff members put on PPE (4).
Health and care workers	Health and care workers is not a separate term, but rather a combination of "health workers" and "care workers" (2).
Health workers	Health workers are all people primarily engaged in actions with the primary intent of enhancing health. This document uses the phrase "health and care workers" to represent both health workers and care workers (2).
High-risk area	The high-risk area is a designated section within the treatment centre allocated for activities and functions that involve direct exposure to the pathogens. This area includes the patients' area and is where the presence of the virus is expected.

Intensive care unit	An intensive care unit can be defined as a specialized section dedicated to providing advanced medical care and continuous monitoring for patients suffering from severe and life-threatening manifestations of Ebola or Marburg diseases.
Laboratory service	Laboratory service refers to a dedicated facility designed to perform a wide range of medical diagnostics, including the detection of filovirus using polymerase chain reaction testing, as well as various biochemistry analyses and other diagnostic procedures.
Low-risk area	The low-risk area is a designated section within the treatment centre allocated for activities and functions that involve no direct exposure to the pathogens. This area, also referred to as the staff area, is where the presence of the virus is not expected.
Morgue	Morgue is the designated area within the treatment centre for the temporary storage of deceased patients' bodies pending final disposition (5). This space may also feature a designated observation area that allows family members to safely view the deceased.
Obstetric and delivery room	An obstetric and delivery room refers to a specially designated area where pregnant women suspected or confirmed with Ebola or Marburg diseases receive focused care during labour and delivery.
Offices	Dedicated spaces within the treatment centre designated for administrative and operational staff to perform non-clinical duties essential to the facility's functioning. These areas are structured to support tasks such as data management, coordination of services, and communication without direct exposure to the pathogens (6).
Outdoor area	It is a designated outdoor space for suspect and confirmed patients.
Screening	Screening is the process in which an individual is evaluated to see whether that person meets a standardized case definition (4).
Suspect patients' area	It is a designated section within the treatment centre specifically allocated for the isolation and clinical management of individuals who meet the case definition for a suspected case (3).
Triage	Triage is the process of sorting patients into categories based on the need for time-sensitive treatment using validated tools. Triage identifies those who require immediate medical intervention, and those who can safely wait. Triage may occur at a health post, primary health centre, clinic, or emergency unit. It typically requires close physical contact (within 1 metre) with the patient during the assessment (4).
Visitor area	It is a designated space within or outside the treatment centre designed to accommodate visitors, allowing for controlled interaction with patients.
Waiting area	The place where patients arrived at the treatment centre and wait to be taken care of by health personnel (1).
Waste area	Waste area is a specifically allocated space within the treatment centre dedicated to storage and management of medical and non-medical waste generated by the facility's operations (7).
WHO-integrate	An evidence-to-decision framework intended to ensure that all criteria of relevance to a health decision are systematically considered (8).

Note: In the operational context of this document, screening and clinical assessment represent distinct physical spaces and actions requiring specific infrastructure and design specifications. Triage, by contrast, is a process rather than a dedicated space. Triage may occur in the waiting area when multiple patients are present, to prioritize the order in which patients access screening. This triage process may include a portion of clinical assessment (such as vital signs measurement or basic symptom severity evaluation) to determine acuity level and urgency. Screening occurs at the screening station, where individual patients are evaluated against the specific case definition to determine if they are suspected or non-cases. Clinical assessment may occur during or after admission, for patients who meet the case definition for suspected cases, to evaluate their clinical status comprehensively, determine severity, and guide treatment decisions.

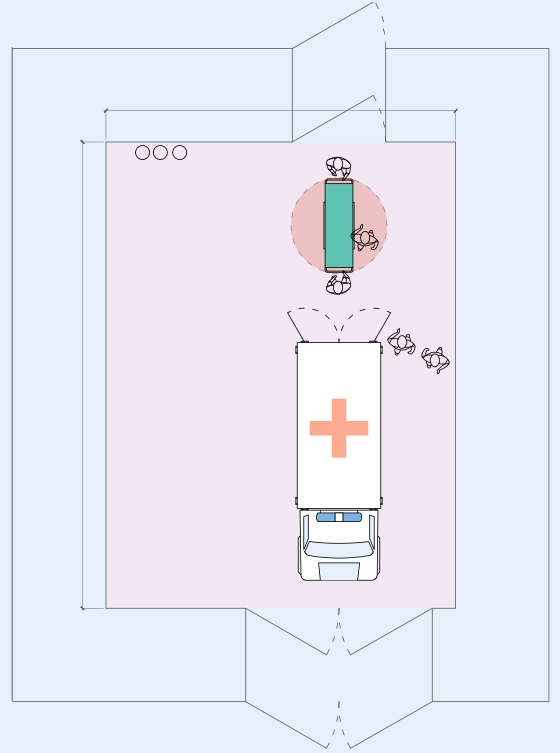
This document provides detailed spatial and infrastructural specifications for screening and clinical assessment, as these represent distinct actions requiring dedicated physical spaces, while triage is recognized as a process.

Executive summary

Filovirus disease outbreaks, caused by Orthoebolavirus and Orthomarburgvirus, continue to pose significant public health threats. Treatment centres have become essential infrastructure for outbreak response, yet they continue to be built without harmonized, evidence-based technical standards. This critical gap has resulted in widely divergent facility designs that compromise staff safety, patient outcomes, and community trust. This document addresses this gap by establishing comprehensive, consensus-driven, evidence-informed technical specifications that every Ebola and Marburg treatment centre must satisfy.

The specifications apply to temporary, semi-permanent, and permanent treatment centres for suspected or confirmed filovirus cases. They address spatial design including zoning and circulation patterns, infrastructure for over 25 functional spaces ranging from screening areas and patient wards to donning and doffing zones, morgues, laundry facilities, waste management, and laboratories. The guidance is intended for facility designers, planners, logisticians, health managers, infection prevention specialists, policymakers, donors, and regulatory bodies involved in planning or operating filovirus treatment centres.

Introduction



Background

The *Filoviridae* family includes two genera, *Orthoebolavirus* and *Orthomarburgvirus* (9), both of which have caused numerous outbreaks with high fatality rates over recent decades (10). Marburg virus was first identified in 1967 during outbreaks in Germany and the former Yugoslavia, linked to infected nonhuman primates imported from Uganda. Since then, it has emerged sporadically across east, central, and south Africa, causing over 14 outbreaks, including major epidemics in the Democratic Republic of the Congo (1998–2000) and Angola (2004–2005) (11). Recently, new outbreaks have been reported in Guinea (2021), Ghana (2022), Equatorial Guinea (2023), Rwanda (2024), the United Republic of Tanzania (2025) and Ethiopia (2025) (12).

Orthoebolavirus were first identified in 1976 during concurrent outbreaks in Zaire (now the Democratic Republic of the Congo) and in the southern part of former Sudan (now South Sudan). Since then, *Orthoebolavirus* have caused more than 40 outbreaks across sub-Saharan Africa with the 2013–2016 west African epidemic resulting in 28 652 cases and 11 325 deaths (11).

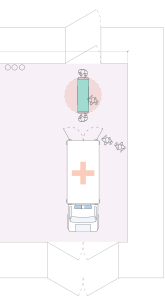
In addition to their structural and genetic similarities, the diseases caused by these viruses, Ebola disease and Marburg disease, also exhibit similar clinical manifestations (13). Once introduced into human populations, both viruses can be transmitted from person to person through direct or indirect contact with the body fluids of an infected individual (4).

Ebola and Marburg treatment centres play a crucial role in responding to these outbreaks by providing a safe environment for patient care and reducing transmission risks through the isolation of infectious patients (14,15). Furthermore, the need for designated treatment centres closer to areas of transmission has been emphasized to enhance the quality of care for patients and mitigate transmission risks (16) at both the facility and community level.

The western Africa Ebola outbreak of 2013–2016 and the North Kivu and Ituri outbreak in the Democratic Republic of the Congo in 2018–2019 accounted for over 90% of the cumulative Ebola virus disease cases since 1976 (17). The unprecedented scale and duration of these outbreaks led to a surge in scientific research, leading to the development of specific vaccines (18), novel experimental treatments (19) and international guidance for optimized supportive care (20).

Despite considerable advances in clinical management (21, 22), there remains a critical gap in the provision of safe, fit-for-purpose infrastructure for filovirus care. Treatment centres continue to be erected without harmonized, evidence-based technical specifications or agreed-upon minimum requirements, resulting in widely divergent designs (23) that compromise the delivery of high-quality, patient-centred and humanized care. Inconsistent zoning, variable utility provisions and ad hoc layouts not only elevate transmission risks for health and care workers (HCWs) and patients but also limit accessibility for families and communities, undermining trust and continuity of care.

To address these deficiencies, this document articulates consensus-driven technical specifications that such centres must satisfy for outbreak response deployment. The following chapter defines the document's aim, delineates its scope and identifies its intended audience, thereby laying the groundwork for a universally applicable standard in filovirus treatment-centre design.



Aim, scope and target audience

This document establishes clear, evidence-informed technical specifications that define the minimum requirements for the design, construction, and operation of treatment centres for the care of patients with suspected or confirmed Ebola and Marburg disease. By articulating mandatory thresholds and complementary recommended features, the guidance seeks to ensure that every facility provides a uniformly safe, functional and patient-centred environment for both HCWs, patients and affected communities.

These specifications apply to all temporary, semi-permanent, and permanent treatment centres, whether newly constructed or adapted from existing structures, that are intended for the isolation and care of patients with suspected or confirmed Ebola or Marburg disease, in both preparedness and response contexts. The document covers spatial zoning and circulation; structural materials and modularity; water, sanitation and waste-management systems; power, ventilation and environmental controls; and considerations for accessibility, cultural adaptability and community engagement.

This document is not intended to replace or duplicate existing World Health Organization (WHO) technical guidance on water, sanitation and waste-management systems, and laboratory requirement. Readers are encouraged to consult WHO publications and guidelines that address water, sanitation and hygiene (WASH) infrastructure, laboratory biosafety standards, waste management technologies, and related technical domains. A list of key complementary documents is provided in the Bibliography to support implementation.

This document serves as a resource for a range of professionals such as health and care workers, health management and support personnel, health facility designers and planners, logisticians, water and sanitation experts, and infection prevention and control (IPC) specialists tasked with planning, funding, designing and operating Ebola and Marburg treatment centres. It may also inform donors, policymakers and regulatory bodies involved in certifying or accrediting filovirus care facilities.

Methods

The development of these technical specifications followed a five-phase, mixed-methods approach designed to ensure both comprehensive evidence synthesis and expert consensus. First, a structured scoping review systematically identified and extracted all relevant literature on the infrastructural design and operational standards of Ebola and Marburg treatment centres. Second, a broad stakeholder consultation, including frontline practitioners, technical specialists and community representatives, prioritized system-level topics and functional areas via an online questionnaire. Third, a detailed morphological analysis of treatment centres built in the last 10 years was conducted to quantify space allocations and identify design evolution over time. Fourth, a targeted expert questionnaire translated these findings into draft “minimum requirements” and “recommended features,” which were appraised against the WHO-INTEGRATE evidence-to-decision framework to ensure that all criteria of relevance to health decision were systematically considered (8). Finally, the methodology, evidence synthesis, technical specifications, including minimum requirements and recommended features statements, were subject to a public consultation process to gather feedback from a broader audience. Following the public consultation, the document was finalized based on the feedback received. Between each phase, meetings were convened with the WHO Secretariat and external experts. These discussions facilitated the exchange of contextual expertise and informed the adaptation of the document to different operational settings. Detailed procedures for each phase are described below.

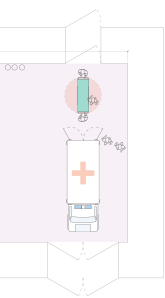
Scoping review

A structured scoping review was undertaken to gather and assess all relevant evidence and practices concerning the infrastructural design of Ebola and Marburg treatment centres and their association with safety and effectiveness. The review was guided by the question: *What are the existing technical specifications, standards and best practices for the design, construction and operation of treatment centres for Ebola and Marburg diseases, and how do these influence morbidity and mortality outcomes during outbreaks?*

Studies were eligible if they focused explicitly on Ebola or Marburg treatment centres, including design, operation, standards, guidelines and technical specifications, and were published in English between January 1995 and July 2024. Peer-reviewed articles, official guidelines, standards documents, technical reports, conference proceedings and government or nongovernmental organization publications were included, provided they took the form of qualitative or quantitative research, reviews, case studies or expert opinion offering insights into construction and operational standards. Records that did not address infrastructure (for example, those limited to clinical treatment protocols), non-English publications, unpublished theses or dissertations, opinion pieces lacking clear author expertise, and duplicate studies were excluded.

Electronic searches were performed in PubMed, Embase and Google Scholar using the following strategies. In PubMed, the query combined (“Ebola” OR “Marburg”) with (“treatment centers” OR “treatment units” OR “treatment facility” OR “isolation center” OR “isolation units” OR “isolation facility”) and the term “structure”. Embase searches employed (‘Ebola’/exp OR Ebola OR Marburg) AND ((treatment AND (‘center’/exp OR center) OR ‘isolation’/exp OR isolation) AND (‘unit’/exp OR unit) OR facility) AND (‘structure’/exp OR structure). Google Scholar searches used the terms Ebola infrastructure AND Marburg infrastructure.

Retrieved records were imported into a reference management system for automated deduplication, followed by manual review. Two independent reviewers screened titles and abstracts against the predefined criteria, with disagreements resolved by consensus. Full-text review was then conducted in duplicate to confirm eligibility. Key data elements, including bibliographic details, facility type and capacity, spatial zoning, dimensions of critical areas, structural materials, ventilation and waste-management systems, utility requirements, and any associations with safety metrics, staff workflow or patient outcomes, were extracted into a standardized charting form. Results are available in [Annex 1](#).



Stakeholder consultation

A structured online questionnaire was distributed to 65 stakeholders representing a balance of technical experts, frontline practitioners and affected community members across WHO regions. Participants were asked to self-identify their primary role (i.e., Ebola/Marburg disease expert, end-user, technical specialist, community representative) and area of expertise (i.e., case management; logistics/WASH; IPC; coordination; laboratory; risk communication; other). The questionnaire then invited respondents to select, from a predefined list, the system-level topics (e.g., site selection, ground characteristics, bed capacity, structural requirements, water/waste management, energy and ventilation, modularity, inclusiveness, cultural adaptability, among others) and the specific services or functional areas (i.e. ambulance bay; waiting area; screening and clinical assessment; donning; doffing; suspect ward; confirm ward; outdoor area; visitor area; morgue; laboratory; convalescent and specialised care spaces; offices; staff and patient sanitation; waste zone) they considered essential for inclusion in the normative guidance. Free-text fields captured additional considerations not covered by the multiple-choice options. Responses were aggregated and analysed descriptively to identify areas of consensus and emerging priorities. Results are available in [Annex 2](#).

Morphological analysis

To gather comprehensive data on the design of Ebola and Marburg treatment centres, members of INITIATE² project (24) were consulted. INITIATE² is a collaborative initiative of WHO and the World Food Programme, bringing together emergency response actors, research institutions, and academic bodies to develop innovative and standardized solutions and trainings to enhance readiness and response capabilities in outbreak situations (24). Participants from INITIATE² were invited to share technical drawings, reports, and photographs of treatment centres constructed by their respective organizations over the past decade. To maximize the number of layouts in the analysis, a systematic search was performed in the MEDLINE, Embase, and Google Scholar databases on the 14th of January 2024, using free-text terms reflecting the eligibility criteria. The search was adapted for each database with the detailed search strategy and eligibility criteria available in the published article (23)). Two authors independently assessed the retrieved references against the eligibility criteria and performed data extraction. All materials received through the consultation and retrieved through literature review were reproduced using computer-aided design software. A unique ID was provided for each treatment centre and standardized coding scheme was developed to ensure consistency across all layouts. Detailed categorization allowed for precise quantification of space allocation and spatial relationships within the centres. In cases of doubt, the provider institution and authors were consulted for clarification. To understand the evolution of design over time, the layouts were divided into three periods: 2014–2017, 2018–2020, and 2021–2023.

A reverse engineering technique (25) was employed to systematically analyse the technical drawings of treatment centres and extract key design and functionality metrics. Reverse engineering involves deconstructing existing systems, in this case, treatment centre layouts, to understand how their design evolved over time and to identify critical elements that contribute to their effectiveness. This approach allowed for the assessment of the spatial dynamics of the centres and how they adapted to the challenges posed by managing infectious diseases. A data extraction tool was developed to extract relevant information pertaining to the design and functionality of the treatment centres according to specific spaces as defined in the glossary. Key metrics included bed capacities, area allocation per functional space and patient, and the ratio of different zones (high-risk, low-risk, outdoor, etc.). Accessibility was evaluated based on standards for healthcare facilities (26) and focused on features like ramps, wide doorways, and navigability for individuals with disabilities in key areas such as waiting rooms, screening, clinical assessment areas, and patient rooms. The treatment centres were categorized as follows based on the construction method used:

- Temporary: Centres set-up with tents.
- Semi-permanent: Centres built with a combination of concrete slabs for flooring, timber for the structural elements, metal corrugated roofing for protection against the elements, and plastic sheeting for vertical enclosures.
- Permanent: Existing buildings repurposed for the outbreak response.

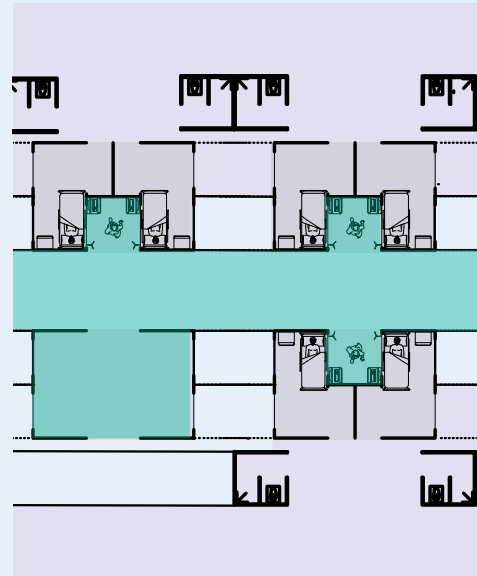
For selected areas, the analysis assessed the level of air and natural daylight exposure using the window-to-floor ratio as a proxy indicator (27). This ratio was calculated by comparing the surface area of openings (such as doors and windows) to the floor surface area. Natural daylight and ventilation levels were considered satisfactory when the window-to-floor ratio was $\geq 12\%$ or $1/8$ (27). To ensure robustness, a subset of the layouts and analyses were independently reviewed by external experts in healthcare architecture and infectious prevention and control. Feedback from these experts was incorporated into the final analysis to enhance the reliability and validity of the results. Complete methods, results and conclusion have been made publicly available through a peer-reviewed publication (23). The results of the morphological analysis have then been shared through an online questionnaire with the expert group to be assessed and synthesized according to the WHO-INTEGRATE evidence-to-decision framework.

Statement formulation

A final, web-based questionnaire presented each of the twenty-five architectural spaces and engineering features identified during the scoping review and stakeholder hearing as a two-fold morphological exercise. First, features such as waiting areas or waste-disposal zones were posed as classification questions, asking experts to indicate whether each should constitute a minimum requirement (i.e., non-negotiable) or a recommended feature (i.e., best practice). Second, dimensional features (e.g., 16 m² donning), bed capacity (e.g. 40 beds), ambulance-bay surface area, surface per patient and so forth, were posed as numeric questions, with respondents selecting from a fixed set of plausible thresholds based on the results of the morphological analysis with the option to provide an alternative answer. For classification questions, every response was tallied and each feature was assigned to “minimum requirement” or “recommended feature” according to a majority vote. In cases where no option attracted more than 50% support, the more conservative stance (“minimum requirement”) was adopted to safeguard safety and functionality.

Dimensional questions were analysed in two passes. The most frequently selected numeric value, the mode, was taken as the recommended feature, reflecting empirical consensus about an optimal design. The second most frequent value was taken as the minimum requirement, establishing a baseline that expert opinion deemed still acceptable under constrained conditions. When two values tied for first place, the lower value was assigned as the minimum requirement and the higher as the recommended feature, thereby setting a baseline for safety without obscuring the aspirational standard (e.g. room dimension for confirmed patients: 9 m² is the minimum standard with the 48% of preference; 12 m² is the recommended feature, with 52% of votes). Each resulting pair of statements, one minimum requirement, one recommended feature, was then reviewed against the WHO-INTEGRATE evidence-to-decision criteria. Feasibility, resource implications, equity and acceptability considerations, as articulated in free-text comments, were woven into the final rationales.

Minimum standards and recommended features



Minimum standards and recommended features

This section provides the minimum standards and the recommended features for the following areas of an Ebola or Marburg treatment centre.

Maximum bed capacity

Ambulance bay

Waiting area

Screening and clinical assessment

Discharge shower

Donning

Doffing

Suspected patient's room

Confirmed patient's rooms

Outdoor area for patients

Visitor area

Intensive care unit

Obstetric and delivery room

Laboratory

Morgue

Spaces for psychosocial support

Children and family spaces

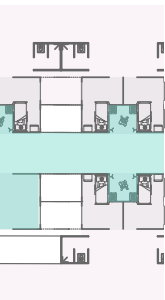
Survivors' clinic

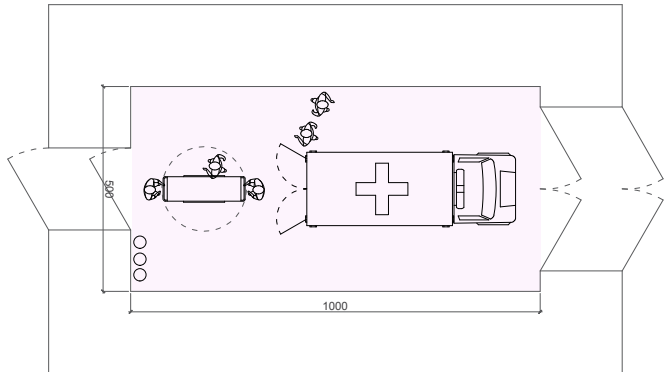
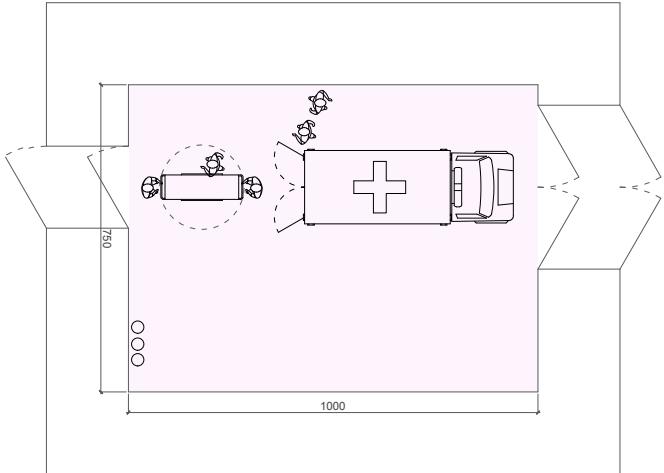
Accessibility

Waste area

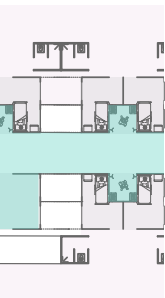
Laundry

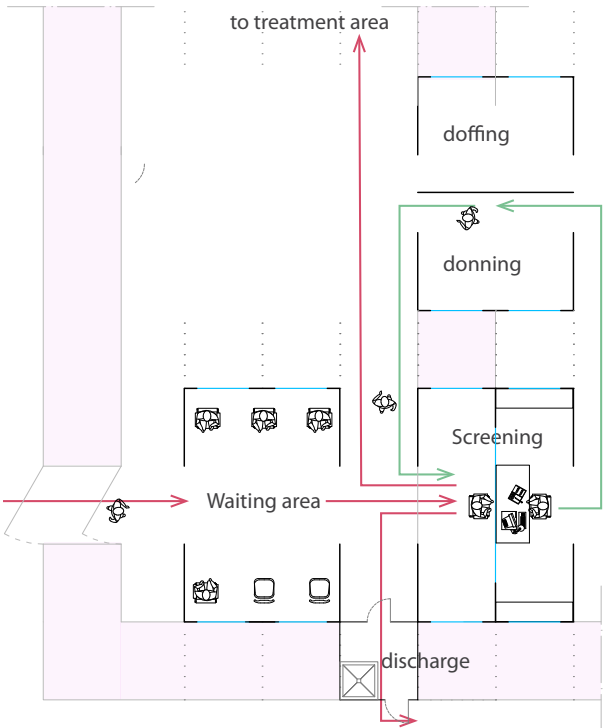
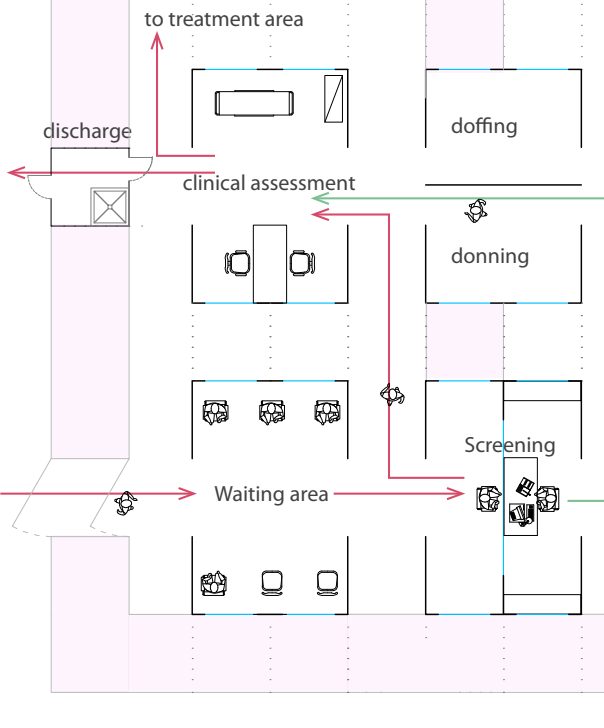
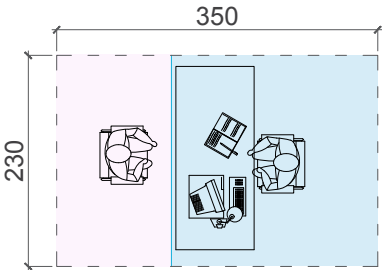
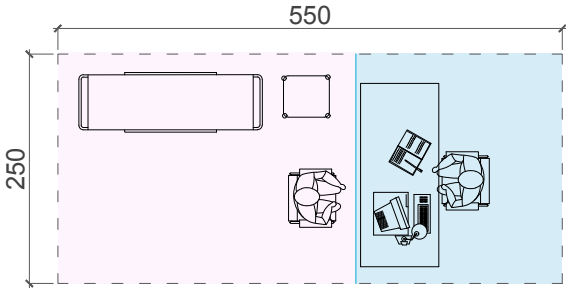
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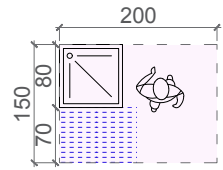
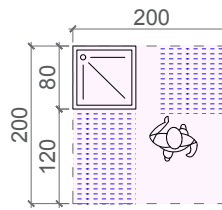


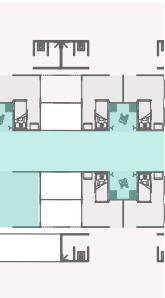
Areas and features	Minimum requirement	Recommended feature	Rationale
Maximum bed capacity (overall capacity including suspects and confirmed cases)	Treatment centres must not exceed a capacity of 40 beds.	It is recommended that centres be limited to no more than 30 beds to ensure optimal safety, functionality, and scalability.	Expert consensus emphasizes that smaller treatment centres are safer and easier to manage, improving outcomes for both patients and staff. Capping capacity at 40 beds, as a mandatory requirement avoids the operational, logistical, and IPC challenges often associated with larger facilities. A 30-bed configuration is recommended as the ideal standard, supporting modular designs that can be rapidly scaled up or down depending on case load and resource availability. This approach aligns with best practices and facilitates a decentralized response, favouring the deployment of multiple smaller centres closer to affected populations, thereby enhancing responsiveness, safety, and sustainability Voting supported this approach. Thirty and 40 beds were the two most-supported sizes (26% each), but preference skewed smaller overall: 41% favoured ≤30 beds (15% for 20, 26% for 30) versus only 18% for larger centres (11% for 50, 7% for 60). A further 15% noted that the maximum depends on contextual factors, and open responses consistently favoured smaller centres. On this basis, 30 beds is adopted as the recommended optimum, while 40 beds is set as an upper limit accommodating those who supported up to 40 while excluding the larger 50–60-bed facilities that few endorsed.
Ambulance bay	A minimum surface area of 50 m² per ambulance must be provided to ensure safe operations. 	A recommended area of 75 m² or more is advised when space permits, to accommodate simultaneous access for two ambulances. 	50 m² is broadly accepted as the operational minimum per ambulance, supporting essential functions. 75 m² provides increased flexibility in high-volume or complex operational contexts. This is a context-sensitive recommendation, allowing adaptation to site constraints while maintaining general occupational health and safety and IPC standards. Key priorities include safe vehicle manoeuvring, safe stretcher handling, staff mobility around the vehicle, and prevention of cross-contamination between clean and dirty zones. Most experts indicated a preference for an area of approximately 50 m² (36%) or 75 m² (32%), which was considered sufficient to allow unobstructed movement and access around the ambulance. A smaller proportion of experts (18%) recommended a larger area of 100 m² to accommodate the presence of two ambulances simultaneously, as highlighted in the open-ended responses.
	- When feasible, surfaces should be flat, non-slippery, and easy to clean. The surface should ensure good wastewater drainage, avoiding water stagnation. - The ambulance bay should be located as close as possible to the screening area to minimize transfer delays.		

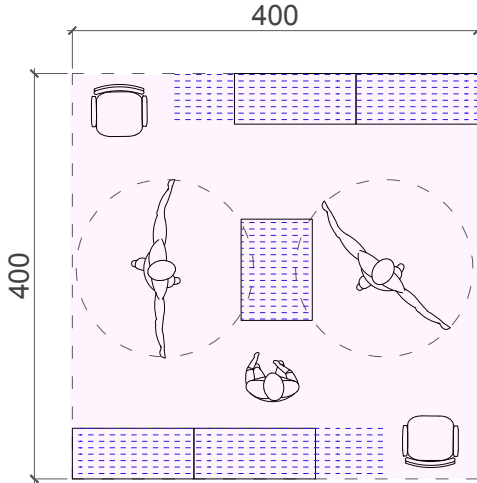
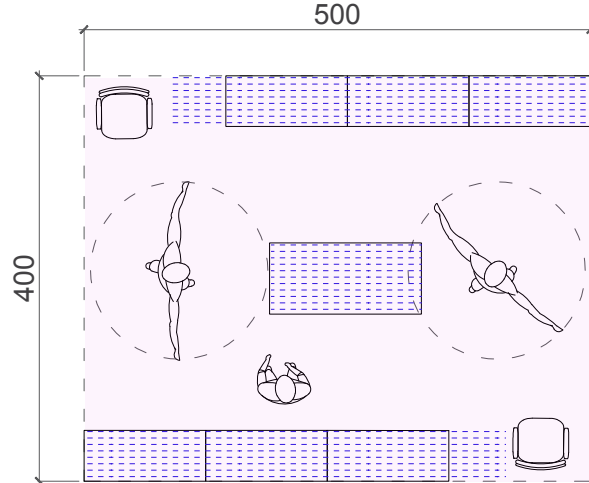
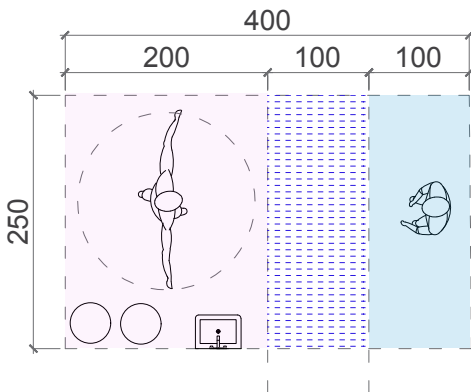
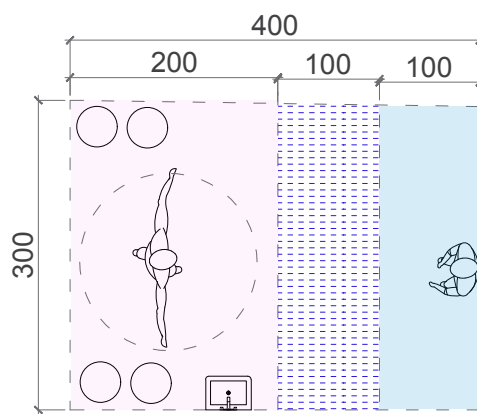
Areas and features	Minimum requirement	Recommended feature	Rationale
Waiting area	The waiting area is a recommended space, not a minimum requirement, in Ebola and Marburg treatment centres. The presence of a waiting area depends on the centre dimensions, the epidemiological scenario, and the specific context.		Waiting area improves patient flow and provide comfort and privacy. However, it should not be mandatory in all contexts due to: - Varying facility layouts and outbreak dynamics - Small centres or those admitting just confirmed patients may not require them - Flexibility is essential for rapid deployment in low-resource settings - Oversized screening zones may already fulfil the waiting function 56% of experts indicated that the waiting area should be designated as a recommended space rather than as a minimum requirement (44%).
	The decision to integrate (A) or detach (B) the waiting area from the screening area should be made based on local conditions, including staff availability, space constraints, and type of construction.		Both integrated and detached configurations can be safe and effective if well managed. Integration may support better use of limited staff and space, while maintaining visibility over patients and enabling quick reorientation. Detached areas, on the other hand, offer clearer physical separation that can enhance infection prevention and ensure patient privacy during screening. Given the wide variability in treatment centre settings, such as tented structures, staffing models, and community expectations, a flexible, context-driven approach is needed. This ensures both safety and operational efficiency while allowing innovative design solutions like visual barriers to adapt to constraints without compromising care quality. 50% of experts supported the statement “The decision should be taken according to the local condition in term of staff availability, surface availability and type of construction”, while 32% supported the integrated approach to ensure visibility toward the waiting zone.
	The minimum surface area per patient in the waiting area is 1.9 m²		A minimum of 1.9 m² per patient ensures safe distancing, movement space, and basic IPC compliance.
	The recommended surface area per patient in the waiting area is 2.5 m²		A recommended 2.5 m² per patient allows for greater comfort, privacy, improved staff workflow, and reduced risk of contamination. Defining both values allows planners to adapt based on space and resources while maintaining safety and efficiency. To ensure effective distancing between patients, most experts favoured an allocation of either 2.5 m² (38%) or 1.9 m² (45%), while 11% reported being uncertain.
	Separate toilet facilities for male and females in the waiting area of treatment centres is a minimum requirement.		Toilets in waiting areas are essential for preserving patient dignity, ensuring basic sanitary needs, and reducing the risk of environmental contamination. Patients often arrive after long journeys and may experience urgent needs, especially in outbreaks involving gastrointestinal symptoms. While cleaning and IPC management pose challenges, these can be addressed through proper staffing and protocols. The benefits of including toilets outweigh the logistical constraints, and their absence could compromise both patient care and outbreak control efforts. 69% of experts identified toilets in the waiting area as a minimum requirement.

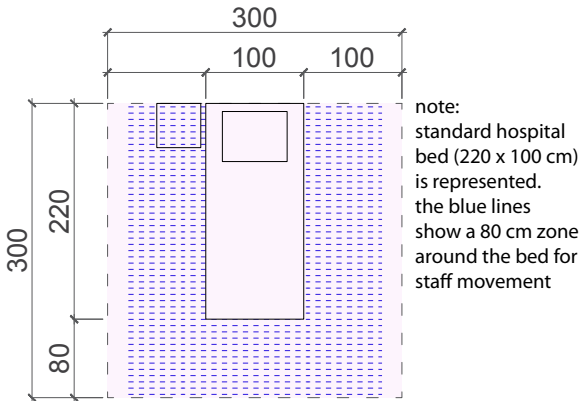
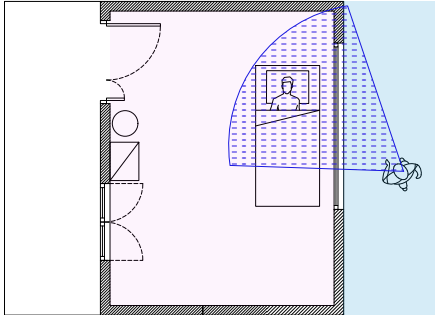
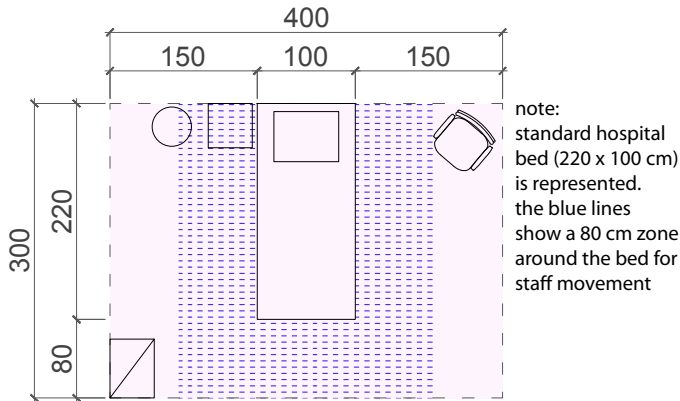


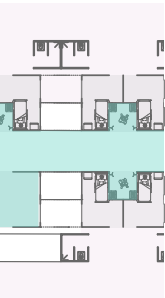
Areas and features	Minimum requirement	Recommended feature	Rationale
Screening and clinical assessment	The decision to locate screening and clinical assessment in the same (A) or separate (B) physical spaces should be based on the specific conditions of each treatment centre, including available space, patient volume, staffing, and IPC requirements. option A: same physical space 	option B: separated space 	Expert input highlights that while screening and clinical assessment serve distinct purposes, with screening requiring no contact and clinical assessment involving close contact, the optimal spatial arrangement depends heavily on local constraints and priorities. Separating the two functions can reduce the risk of cross-contamination and clarify workflow, especially given their differing IPC protocols. However, integrating them may support rapid assessment of critically ill patients, reduce human resource needs, and streamline patient flow in low-resource settings. Given the wide variation in treatment centre layouts, resource availability, and outbreak dynamics, flexible guidance is needed. Treatment centres should be empowered to adapt their design while ensuring safe, efficient, and context-appropriate patient management. 38% of experts indicated that “the decision to locate screening and clinical assessment in the same (A) or in separate (B) physical spaces should be based on the specific conditions of each treatment centre, including available space, patient volume, staffing, and infection prevention and control (IPC) requirements.” By contrast, 31% preferred the integrated option, while 28% favoured the opposite approach, with screening and clinical assessment fully separated.
	A minimum of 8 m² per screening station is required in all Ebola and Marburg treatment centres. Where a transparent screen separates the HCWs and patient, this 8 m² threshold remains sufficient. If separation is achieved using double fences with a 2 metres wide buffer between them, add an additional 4 m² (for a total of 12 m² per station) to accommodate the buffer zone. 	In settings where clinical assessment (involving close physical contact and PPE use) occurs in the same space, up to 14 m² per station is recommended. With a transparent screen in place, 14m² continues to ensure safe workflows. When using a double-fence separation with a 2 metres buffer, increase the area by 4 m² (for a total of 18 m²) to maintain unimpeded movement and effective IPC controls. 	Expert consensus supports 8 m² as the minimum area required to ensure safe and effective screening. This space allows for a chair in the patient zone, a table and chair in the staff zone, and the installation of a transparent barrier or a double-fence separation with a 2 metres distance to maintain visual contact while minimizing infection risk. In settings where clinical assessment (involving physical contact and use of PPE) is also performed in the same space, up to 14 m² may be needed to accommodate movement, equipment, and patient transport devices such as stretchers. Experts emphasize the need for flexibility in design but also note that too little space may lead to fatigue, errors, and compromised infection prevention practices. Nearly half of respondents (47%) indicated a preference for 8 m², accommodating two chairs with a table in between and a transparent screen. A smaller proportion (25%) preferred 4 m², allowing only two chairs and a transparent screen, while 21% favoured 14 m² to include a stretcher for clinical assessment in the high-risk area.

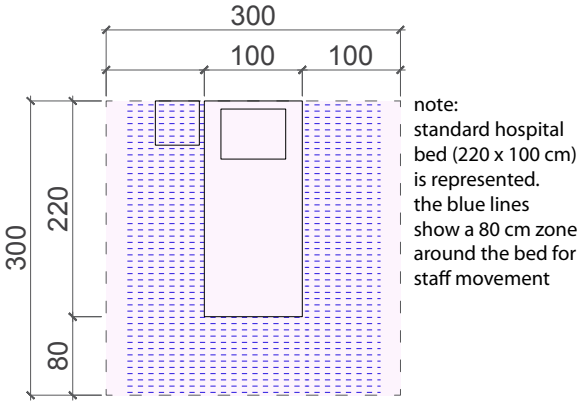
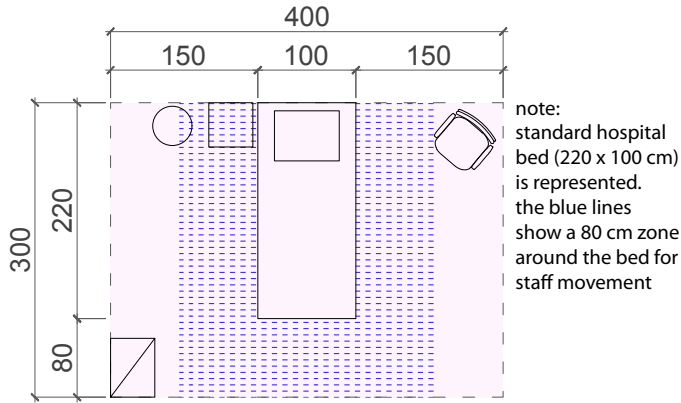
Areas and features	Minimum requirement	Recommended feature	Rationale
Discharge shower	A minimum of 3 m² is required for each discharge shower in Ebola or Marburg treatment centres. 	4 m² is recommended when space and resources allow, to enhance comfort, workflow, and patient safety. 	Expert consensus supports 3 m² as a practical minimum that balances IPC, patient privacy, and operational feasibility. This space allows for unidirectional flow, safe movement, and the ability for patients to undress, shower, and change clothes without compromising dignity or hygiene. While larger spaces (4 m²) may be beneficial in contexts requiring staff assistance or greater accessibility, 3 m² was viewed as sufficient in most cases, especially in low-resource settings where space and infrastructure are limited. The design should also ensure ease of cleaning and adherence to IPC protocols (e.g. washable surfaces, floor drainage), making 3 m² a reasonable and adaptable baseline. Expert responses were distributed across several options. However, 3 m² emerged as the preferred choice, receiving 36% of the votes, followed by 4 m² (25%), 2 m² (21%), and 5 m² (11%).
		Discharge showers located near the screening and clinical assessment area should be considered a recommended space, not a minimum requirement, in Ebola or Marburg treatment centres.	While discharge showers can contribute to infection prevention and patient dignity, expert feedback indicates that their necessity depends on the specific design and practices of each centre. When appropriate IPC measures are in place, such as physical distancing, environmental cleaning, and hand hygiene, the risk of contamination for patients not meeting the case definition is minimal. Implementing discharge showers requires additional infrastructure, supplies, and staffing, which may not be justifiable given their limited use in many settings. A flexible approach allows treatment centres to prioritize resources where the risk of exposure is higher, while still maintaining safe and effective patient flow. Most experts (57%) agreed that the discharge shower located near the screening area should be considered a ‘recommended space’, while 36% supported its designation as a ‘minimum requirement’. Only 7% suggested ‘other’ options.
Donning	A ratio of one donning area per 15 beds is required in Ebola or Marburg treatment centres. This ratio represents the upper limit of what is considered acceptable, particularly in smaller or resource-limited facilities. Adjustments should be made based on facility size, layout, patient flow, and staffing capacity.	A ratio of one donning area per 10 beds is recommended to minimize the risk of cross-contamination, reduce staff waiting times, and facilitate more efficient rotation and movement. This lower ratio is ideal for ensuring safety and maintaining workflow efficiency, particularly in high-volume or high-risk settings.	Expert consensus supports a 10–15 bed-to-donning area ratio as a balanced approach that ensures both safety and practicality. Lower ratios (e.g., 1:10) help prevent overcrowding, minimize the risk of cross-contamination, and improve adherence to PPE protocols. They also facilitate quicker staff rotation and reduce walking distances between donning areas and patient zones, which is critical during high workload or emergency situations. While 1:10 is ideal for efficiency and infection prevention, 1:15 may be acceptable in smaller or resource-limited centres. Experts emphasized that the optimal ratio should be context-specific, considering the number of HCWs, patient flow, available space, and logistical constraints. Flexibility in implementation ensures that centres can maintain safe and functional workflows under varying conditions. Most responses were distributed between one donning area for every 10 beds (33%) and one donning area for every 15 beds (30%). Fewer respondents preferred other options, with 19% supporting one donning area for every 20 beds and 7% suggesting a single donning area per centre. In the open responses, 11% of experts emphasized that the required number of donning areas depends on contextual factors such as layout, staff numbers, and circuit design.

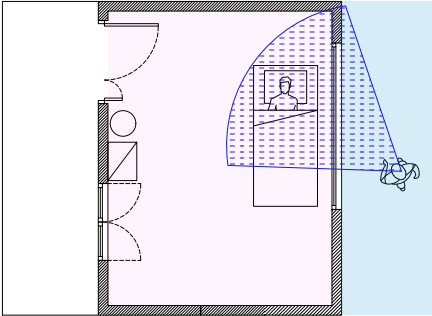
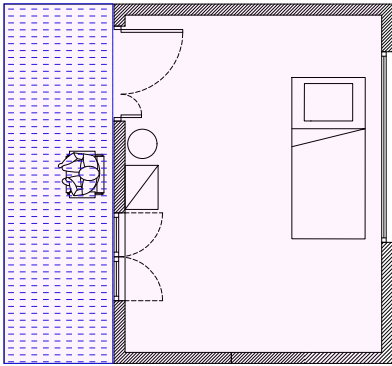


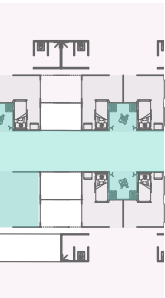
Areas and features	Minimum requirement	Recommended feature	Rationale
	A minimum of 16 m² is required for each donning area in Ebola or Marburg treatment centres, accommodating at least two staff members donning PPE simultaneously, along with a third person supporting the buddy system. 	Where space and resources allow, 20 m² is recommended for each donning area. This allows full rotation with open arms of two staff members, provides approximately 10 linear metres of shelving for PPE and supplies, and accommodates seating and a central table to facilitate the donning procedure. 	Experts agree that 16 m² provides the necessary space for safe, efficient, and comfortable donning of PPE. This surface allows two staff to move freely, bending, turning, and stretching. It also accommodates the essential buddy system, which requires a third person to observe and verify correct PPE use. Additional space is needed for PPE storage, hand hygiene stations, mirrors, bins for waste, and a small table or surface and chairs/bench to facilitate the donning process. While some flexibility may be applied depending on available resources and layout constraints, allocating less space risks hindering movement, reducing compliance with protocols, and increasing infection risk. 41% of experts identified 16 m² as the minimum requirement, while 29% supported 8 m². Smaller proportions voted for 20 m² (15%) and 9 m² (11%).
Doffing	A minimum of two doffing lines is required in every doffing station. A ratio of additional one doffing line per 10 beds is required in Ebola or Marburg treatment centres. This ensures that health and care workers can safely remove PPE without delay, particularly during high-demand periods such as shift changes or emergencies. Adjustments should be made based on facility layout, staff workload, and patient flow. A minimum of 10 m² is required per doffing line in Ebola or Marburg treatment centres. 	A minimum of two doffing lines is recommended in every doffing station. A ratio of additional one doffing line per 5 beds is recommended to minimize overcrowding and reduce the risk of self-contamination during PPE removal. This lower ratio supports safer, more supervised doffing procedures, especially in high-risk or high-temperature environments where fatigue and stress may impair staff performance. A surface of 12 m² is recommended when space and resources allow, to enhance comfort, workflow, and safety. 	Expert consensus emphasizes that doffing is a high-risk procedure requiring time, space, and supervision to minimize the risk of self-contamination. A ratio of one doffing line per 10 beds, with a minimum of two doffing lines, ensures that HCWs can remove PPE safely and without delay, especially during peak times like shift changes or emergencies. Fatigue, heat stress, and urgency can increase the likelihood of errors during doffing. Increasing the number of doffing lines reduces overcrowding, supports better adherence to infection prevention protocols, and helps maintain consistent standards of care. While this ratio serves as a practical recommendation, treatment centres should adapt the number of doffing stations to their specific operational context. 44% of experts supported one doffing line for every 10 beds as the minimum requirement, while 19% favoured the more stringent option of one doffing line for every 5 beds, and another 19% supported one doffing line for every 20 beds. Experts agree that 10 m² provides the necessary space for HCWs to safely remove PPE. Doffing is a critical step in infection prevention, requiring clear separation between clean and contaminated zones, as well as enough room for full body movement, bending, rotating, and stabilizing, without accidental contact with surfaces or equipment. This space allocation also accommodates essential items such as separated hands-free containers for reusable and disposable PPE, hand hygiene stations, mirrors, seats for fatigued staff, emergency eyewash access, and chlorine solutions. In addition, it allows space for a buddy or support staff to assist in case of complications during doffing. While some contexts may benefit from slightly larger areas (e.g., 12 m²), the 10 m² standard offers a practical and achievable balance between safety, functionality, and spatial efficiency. Most respondents (60%) identified 10 m² as the minimum surface area per doffing line, while 30% supported 12 m² and a smaller proportion (7%) preferred 8 m².

Areas and features	Minimum requirement	Recommended feature	Rationale
		Doffing areas in Ebola or Marburg treatment centres should be designed as open structures with a roof, allowing for ventilation while providing shelter from the elements.	Experts widely agree that open-sided doffing structures provide critical advantages for both infection prevention and staff safety. This design allows for natural ventilation, which helps disperse chlorine odours and reduce heat buildup, especially important in tropical climates where staff are already under physical stress from PPE use. An open structure also ensures clear visibility, enabling effective supervision and adherence to proper doffing protocols through the buddy system. A roof provides necessary protection from sun and rain, preserving the effectiveness of chlorine solutions and ensuring staff comfort. While local weather and space constraints may require adaptation, this open-yet-covered design offers a practical balance between ventilation, visibility, and operational efficiency. The vast majority of experts (89%) selected the open-sided structures as best practice for doffing areas.
Suspected patient's room	Individual rooms with a toilet and shower are required for all suspected patients in Ebola or Marburg treatment centres. A minimum of 9 m² per bed (excluding toilet space) is required for suspect patient rooms in Ebola or Marburg treatment centres.  note: standard hospital bed (220 x 100 cm) is represented. the blue lines show a 80 cm zone around the bed for staff movement Direct visibility into suspected patients' rooms from low-risk areas is required in at least some rooms of Ebola and Marburg treatment centres to enable continuous monitoring and prompt response while minimizing exposure risk. Facilities should prioritize visibility in rooms where continuous observation is most critical (e.g., children, pregnant women, higher acuity patients). 	A surface of 12 m² (excluding toilet space) is recommended when space and resources allow, to enhance comfort, workflow, and safety.  note: standard hospital bed (220 x 100 cm) is represented. the blue lines show a 80 cm zone around the bed for staff movement Direct visibility into all suspect patient rooms is recommended where design and resources allow. This supports continuous monitoring, reduces PPE use, and fosters transparency and patient trust, while implementation methods (e.g., transparent barriers, observation windows, or close-circuit television) may be adapted to the local context.	Expert consensus strongly supports individual rooms as essential to ensure effective IPC and critical to prevent cross-contamination during the diagnostic period. Shared spaces significantly increase the risk of nosocomial transmission. Individual rooms with integrated toilet and shower facilities limit patient movement, reduce environmental contamination, and enhance both patient safety, physical and psycho-social well-being. While implementation may be challenging in certain emergency or resource-constrained settings, establishing this as a minimum standard sets a clear benchmark for safe, dignified, and effective care. Most experts (74%) proposed individual rooms for suspected cases as a minimum standard, whereas 26% considered them a recommended requirement. Expert consensus highlights that 9 m² provides sufficient space to ensure patient comfort, support safe and effective healthcare delivery, and uphold IPC standards. This surface area allows HCWs to move freely around the bed, access both sides, and perform medical procedures without spatial constraints. It also accommodates minimal furnishings and equipment while maintaining clearance for cleaning and disinfection. Larger rooms help reduce feelings of confinement and support a more humanized and dignified care environment, especially important given the psychological and social stress associated with isolation. While a 9 m² space may be acceptable in highly constrained contexts, 12 m² is more appropriate to meet the evolving expectations for quality of care, safety, and operational functionality in treatment centres. Experts were divided between recommending a minimum of 9 m² for an individual room (44%) and a minimum of 12 m² (56%). Experts agree that direct visibility is essential to ensure continuous patient monitoring, particularly for clinically unstable individuals. This design feature enables HCWs to observe patients without entering high-risk areas, thereby reducing unnecessary PPE use, improving response time, and enhancing staff safety. Direct visibility also supports IPC by limiting physical exposure while maintaining supervision. Furthermore, it strengthens patient dignity, allows for basic communication and psychological support, and fosters community trust, especially in high-mortality outbreaks where transparency is critical. While implementation methods may vary, such as transparent barriers or close-circuit television, the requirement for direct visibility must be upheld across all centres. 44% of experts preferred ensuring visibility in all suspected rooms, whereas 30% supported ensuring visibility in only some rooms, and 22% suggested considering visibility as a best practice rather than as a minimum standard.

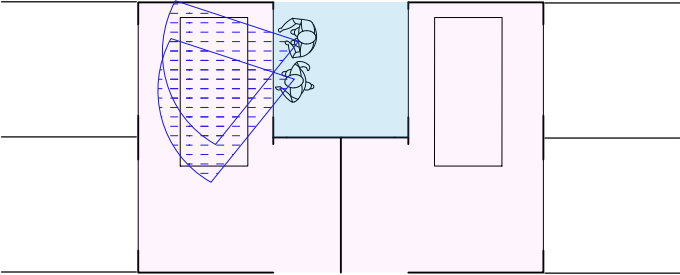


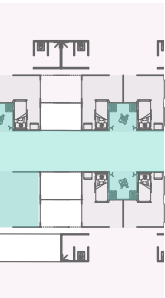
Areas and features	Minimum requirement	Recommended feature	Rationale
	Each suspect patient room in Ebola or Marburg treatment centres must have at least one operable window to provide natural ventilation, light, and visibility toward outside. This is essential to reduce the risk of airborne contaminants accumulation and to support patient comfort and trust.	A window-to-floor ratio of at least 1:8 is recommended to ensure adequate airflow and daylight exposure, enhancing infection prevention and psychological and social well-being. This ratio should be applied where architectural and operational conditions allow, while maintaining flexibility in settings with structural or environmental constraints.	Experts emphasized that windows are critical for ensuring natural ventilation and light, which support both IPC and patient well-being. Adequate ventilation helps reduce the concentration of airborne pathogens, particularly in rooms where coughing or aerosol-generating procedures may occur. Natural light contributes to psychological comfort and improves the therapeutic environment. The 1:8 window-to-floor ratio is recognized as a practical standard to allow for sufficient airflow and daylight exposure. While local conditions (such as rain, maintenance, and mosquito netting) may affect window use, the presence of at least one functional window remains essential. Visual contact with the outside world also fosters transparency and strengthens patient trust in the care process. 44% of experts emphasized that, as a minimum standard, it is essential not only to recommend the presence of windows but also to specify a sizing criterion, such as a window-to-floor ratio of 1:8. In comparison, 26% considered the presence of a window alone sufficient as a minimum standard, while another 26% chose not to provide an opinion.
Confirmed patient's rooms	A minimum of 9 m² per bed (excluding toilet space) is required for confirmed patient rooms in Ebola or Marburg treatment centres. 	It is recommended that confirmed patients in Ebola or Marburg treatment centres be accommodated in individual rooms that include a toilet and shower, where space and resources allow. A surface of 12 m² (excluding toilet space) is recommended when space and resources allow, to enhance comfort, workflow, and safety. 	Individual rooms with dedicated toilets and showers support patient dignity, comfort, and infection prevention by minimizing the risk of co-infections and enhancing hygiene. While confirmed patients pose no transmission risk to one another for Ebola or Marburg, risk of nosocomial infection with other pathogens cannot be ruled out. However, during large outbreaks or in resource-limited settings, shared arrangements may be necessary, and design flexibility should be maintained. The majority of experts (57%) agreed that individual self-contained rooms for confirmed cases should be considered a recommended feature rather than a minimum standard, while 36% supported designating this feature as a minimum standard. Expert consensus highlights that 9 m² provides sufficient space to ensure patient comfort, support safe and effective healthcare delivery, and uphold IPC standards. This surface area allows HCWs to move freely around the bed, access both sides, and perform medical procedures without spatial constraints. It also accommodates minimal furnishings and equipment while maintaining clearance for cleaning and disinfection. Larger rooms help reduce feelings of confinement and support a more humanized and dignified care environment, especially important given the psychological stress associated with isolation. While a 9 m² space may be acceptable in highly constrained contexts, 12 m² is more appropriate to meet the evolving expectations for care quality, safety, and operational functionality in treatment centres. Experts were divided between recommending a minimum of 9 m² for an individual room (48%) and a minimum of 12 m² (52%).

Areas and features	Minimum requirement	Recommended feature	Rationale
	Direct visibility from low-risk areas into at least some confirmed patient rooms is required in Ebola or Marburg treatment centres to enable safe monitoring of clinically unstable patients. Facilities should prioritize visibility in rooms where continuous observation is most critical (e.g., children, pregnant women, higher acuity patients). 	Direct visibility from low-risk areas into all confirmed patient rooms is recommended to strengthen patient monitoring, reduce unnecessary entry into high-risk areas, improve staff safety, and enhance patient dignity and trust.	Experts agree that direct visibility is essential to ensure continuous patient monitoring, particularly for clinically unstable individuals. This design feature enables HCWs to observe patients without entering high-risk areas, thereby reducing unnecessary PPE use, improving response time, and enhancing staff safety. Direct visibility also supports IPC by limiting physical exposure while maintaining supervision. Furthermore, it strengthens patient dignity, allows for basic communication and psychological support, and fosters community trust, especially in high-mortality outbreaks where transparency is critical. While implementation methods may vary, such as transparent barriers or close-circuit television, the requirement for direct visibility must be upheld across all centres. Most respondents (62%) preferred ensuring visibility in all confirmed rooms, whereas 23% supported ensuring visibility in only some rooms, and 15% suggested considering visibility as a best practice rather than as a minimum standard.
	Each suspect patient room in Ebola or Marburg treatment centres must have at least one operable window to provide natural ventilation, light, and visibility toward outside. This is essential to reduce the risk of airborne contaminants accumulation and to support patient comfort and trust.	A window-to-floor ratio of at least 1:8 is recommended to ensure adequate airflow and daylight exposure, enhancing infection prevention and psychological well-being. This ratio should be applied where architectural and operational conditions allow, while maintaining flexibility in settings with structural or environmental constraints.	Experts emphasized that windows are critical for ensuring natural ventilation and light, which support both IPC and patient well-being. Adequate ventilation helps reduce the concentration of airborne pathogens, particularly in rooms where coughing or aerosol-generating procedures may occur. Natural light contributes to psychological comfort and improves the therapeutic environment. The 1:8 window-to-floor ratio is recognized as a practical standard to allow for sufficient airflow and daylight exposure. While local conditions (such as rain, maintenance, and mosquito netting) may affect window use, the presence of at least one functional window remains essential. Visual contact with the outside world also fosters transparency and strengthens patient trust in the care process. Most respondents (61%) supported considering an operable window as the minimum standard for confirmed cases, whereas 15% preferred to designate the window with a 1:8 window-to-floor ratio as the minimum standard.
Outdoor area for patients	An outdoor area is a minimum requirement for Ebola or Marburg treatment centres, as it plays a critical role in patient well-being and recovery. However, the minimum surface area per patient should remain flexible and depend on the total facility capacity and outbreak context. 		Outdoor areas contribute significantly to patients' mental and physical recovery by offering fresh air, natural light, and space for movement, especially during extended stays. Making outdoor space a minimum standard ensures that every patient has access to this vital component of care. At the same time, fixed surface area requirements may not be practical for all contexts. Factors such as available land, outbreak size, and facility design should inform the actual space allocation, allowing facilities to maintain flexibility while still promoting patient-centred environments. Half of the experts (50%) supported designating the outdoor area as a minimum standard, while 43% considered it a recommended feature. Only 7% indicated that outdoor areas are not a priority in a treatment centre.



Areas and features	Minimum requirement	Recommended feature	Rationale
Visitor area	A designated visitor area is a minimum requirement for Ebola or Marburg treatment centres to support patient well-being, ensure transparency, and promote community trust. The visitor area must include a roof to protect from sun and rain and provide enough space per patient to ensure a dignified and comfortable environment. Visitors are required to access the visitor's area without entry into the low-risk (staff) area.	It is recommended that visitor areas be located within 10 metres of patient rooms to enhance patient well-being and support family engagement, provided that access is structured to prevent visitor entry into low-risk (staff) zones and aligned with IPC protocols. Where possible, the visitor space should accommodate multiple family members and be designed with attention to cultural norms and spatial flexibility.	Family presence and interaction play a critical role in patient mental health and emotional support, especially in contexts where isolation can contribute to fear and stigma. Visitor areas foster trust and acceptance of treatment centres within communities, helping to dispel misinformation and improve health-seeking behaviours. A roofed area ensures visitors are protected from the elements, supporting more regular and humane interactions. A minimum surface area allows for adequate distancing, comfort, and cultural appropriateness. Proximity facilitates meaningful contact while maintaining safety through well-designed separation from clinical and staff areas. The majority of experts (82%) emphasized the importance of a designated visitor area as a minimum standard, while a smaller proportion (18%) considered it a recommended feature. 53% recommended that the visitor area be located close to patients, whereas 18% did not express an opinion on the distance, and 11% indicated that the appropriate distance should be flexible, depending on local contextual factors.
Intensive care unit		An intensive care unit should be considered a recommended feature in Ebola or Marburg treatment centres, based on the scale of the outbreak, the specific needs of the patient population, and available resources. In addition, some patient rooms should be designed to be adaptable for intensive care to ensure flexible and scalable response capacity.	Experts emphasized that while intensive care unit care is critical for managing severely ill patients with Ebola or Marburg, establishing a fixed intensive care unit in every centre may not always be feasible due to limitations in resources, infrastructure, and staffing. A recommended approach allows treatment centres to respond proportionally to the severity and scale of an outbreak. Combining dedicated intensive care unit rooms with adaptable patient spaces ensures flexibility, supports effective triage, optimizes care delivery, and enables facilities to manage critical patients safely while respecting design, logistics, and infection prevention constraints. Most respondents (53%) considered the intensive care unit to be a recommended feature rather than a minimum standard (41%), while 6% did not express an opinion.
Obstetric and delivery room	A dedicated obstetric and delivery room is a minimum requirement in Ebola and Marburg treatment centres to ensure safe, specialized care for pregnant patients.		Pregnant women have consistently been affected during outbreaks, with high rates of pregnancy loss and complications. Experts emphasized that these patients require tailored medical attention, privacy, and environments equipped to manage high-risk deliveries. Dedicated spaces ensure appropriate infection control, safe delivery practices, and protection for HCWs. While flexibility may be necessary in emergencies, every treatment centre must be equipped or have a contingency plan to manage obstetric care safely and effectively. Half of respondents (50%) considered the obstetric and delivery room as a minimum standard rather than a recommended space (41%), while 9% did not express an opinion.
Laboratory (excluding polymerase chain reaction)	A laboratory space capable of conducting biochemistry analyses (excluding Ebola and Marburg polymerase chain reaction diagnostics) is required in all Ebola and Marburg treatment centres to ensure comprehensive diagnostics and effective patient management. Details information on the design and requirements for laboratory is available in the <i>Laboratory biosafety manual, 4th edition</i> (28).		Access to biochemistry testing is essential for monitoring organ function, guiding supportive care, and responding to complications such as renal failure or electrolyte imbalances. In addition to improving clinical outcomes, an on-site laboratory facilitates rapid diagnostics and safer handling of biological fluids. While point-of-care tools are useful, a dedicated lab space enhances the quality and continuity of care, particularly in severe or complex cases. Integrating laboratory capacity into treatment centre design is critical to ensure a high standard of patient care during outbreaks. 43% of experts consider the laboratory should be minimum standard, the 39% consider it should be a recommended function, while the 18% did not express an opinion.

Areas and features	Minimum requirement	Recommended feature	Rationale
Morgue	A morgue is a minimum requirement in all Ebola and Marburg treatment centres to ensure the secure and dignified management of deceased patients in alignment with IPC protocols and cultural practices. A minimum of two dead body storage units per 10 beds is required to meet anticipated mortality rates and prevent delays in safe and dignified burials. A designated visitor area within the morgue must also be included to allow families to safely view and pay their respects to the deceased. 		Despite improved treatment options and the availability of vaccines, fatalities remain a significant concern in Ebola and Marburg outbreaks. A baseline morgue capacity is essential even in smaller centres, with the ability to scale up based on bed capacity. This ensures bodies are handled, stored, and prepared in a safe manner that minimizes transmission risk and aligns with IPC standards. Equally important is the provision of a visitor area to allow families to say goodbye in a culturally sensitive, secure, and respectful manner. Short morgue stays are typical, but the facility must always be available to prevent delays in burial. Incorporating cultural, IPC, and anthropological considerations strengthens community trust, promotes acceptance of care, and reduces the risk of misinformation or resistance. Together, the morgue and visitor area uphold dignity, ensure transparency, and bridge operational needs with community expectations. The vast majority of experts (75%) consider the presence of a morgue in an Ebola or Marburg treatment centre a minimum standard, a minority (7%) prefer to consider it a recommended function, while 7% prefer to focus on protocols for managing dead bodies instead of space and 11% did not express an opinion. Last, 72% suggest that the morgue should have a dedicated space for families.
Spaces for psychosocial support		It is recommended that all Ebola and Marburg treatment centres include a designated space or clearly identified provision for psychological and social support services, both for HCWs and for the families of patients. While not a minimum infrastructure requirement, providing flexible and accessible mental health and psychosocial support is essential to address the emotional and psychological burden of working in or interacting with treatment centres during outbreaks.	The psychological toll on both staff and families during Ebola and Marburg outbreaks is profound, with significant implications for mental health, burnout, resilience, and community trust. While not all centres may have the space or resources to set up dedicated psychological offices, the presence of accessible mental health and psychosocial support services is critical. Staff are often exposed to extreme stress and trauma, and families face fear, stigma, grief, and isolation. Providing mental health and psychosocial support, whether through dedicated offices, repurposed spaces, or mobile or virtual services, can enhance staff well-being, reduce burnout, and improve quality of care. For families, it supports emotional resilience, fosters trust in the health system, and aids community understanding and acceptance of outbreak control measures. Flexibility is essential: psychological and social services should be adapted to the size of the centre, intensity of the outbreak, and availability of resources, with services embedded into outbreak response strategies even if dedicated rooms are not feasible. 59% of experts consider the mental health and psychological support service for staff a recommended function, while the 41% consider it should be a minimum standard. Half (48%) consider the mental health and psychological support service for patients' relatives a recommended function, while the other half 48% consider it should be a minimum standard.
Children and family spaces		Ebola and Marburg treatment centres should include a dedicated space or adapted area to support the specific needs of children and families, where feasible. While not a minimum requirement, these spaces should be considered during the design and setup phase based on available resources, the centre's capacity, and the expected demographic of admitted patients.	Children and families affected by Ebola and Marburg disease require particular attention, both for medical and psychosocial support. Dedicated spaces can improve the comfort, safety, and emotional well-being of paediatric patients and their caregivers. However, their provision must be balanced with IPC constraints, available space, and the intensity of the outbreak. Historically, the inclusion of such spaces has declined, raising concerns about the erosion of patient-centred and family-friendly care. While some respondents view these spaces as essential for dignified, humanized care, others caution against the potential IPC risks and operational challenges they may introduce. In contexts where family accompaniment is possible and safe, and where children are admitted, incorporating child-friendly features or family support zones is strongly encouraged. Even if dedicated rooms are not feasible, treatment centres should strive to adapt existing spaces to accommodate paediatric needs and ensure emotional support to families, particularly when entire households are affected. 37% of experts considered a child- and family-friendly space to be a minimum standard, another 37% regarded it as a recommended function, while 26% did not express an opinion.



Areas and features	Minimum requirement	Recommended feature	Rationale
Survivors' clinic	The following spaces are considered minimum requirements in a survivor's clinic: • Screening and registration. • Medical consultation room. • Biological fluids sampling room. • Counselling and psychosocial support room.	Survivor clinics are a recommended feature during Ebola and Marburg outbreaks. While not a minimum requirement (in terms of physical space, not as intervention), they should be considered based on the scale of the outbreak, the number of survivors, and the capacity of the response. It is also recommended that, where possible, survivor clinics be located outside of the treatment centre to reduce stigmatization, support continuity of care, and promote reintegration into the community. The following spaces are considered recommended features in a survivor's clinic: • Delivery room. • Pharmacy. • Warehouse/storage space.	Survivor clinics provide essential follow-up care for individuals recovering from Ebola or Marburg disease, addressing long-term complications, monitoring viral persistence in body fluids, and supporting management of clinical sequelae and psychosocial well-being. These services are vital for survivor resilience, reintegration, and for reducing the risk of onward transmission or secondary outbreaks. However, survivor care often extends over 12 to 24 months and requires integration into the broader health system for sustainability. Locating these clinics within treatment centres may lead to stigmatization and create emotional distress for survivors, especially those who lost loved ones in the same facility. Many respondents emphasized that outpatient care models, delivered in repurposed or existing health facilities, are better suited for long-term support. Thus, while survivor clinics are a critical component of the response, they should be adapted to local needs and settings, with clear transition plans to mainstream health services once the acute phase of the outbreak subsides. A total of 42% of respondents considered the survivor clinic to be a recommended function rather than a minimum standard (27%). 12% did not express an opinion, while 19% selected “other,” with several emphasizing that the survivor clinic should be located outside and separated from the Ebola treatment centre. These categorizations reflect the core services required to ensure safe clinical follow-up and psychosocial support for survivors, while allowing flexibility for additional services based on the context, available space, and expected volume of patients. Regarding survivor clinics, 62% considered screening and registration a minimum standard; 69% considered the medical consultation room a minimum standard, 58% considered the biological fluids sampling room a minimum standard and 58% considered the counselling and psychosocial support room a minimum standard. 27% considered the delivery room a minimum standard, while 23% considered it a recommended function and 31% considered that it should not be included. 35% considered the pharmacy a minimum standard while 31% considered it a recommended space. 46% considered the warehouse a recommended space while 11% considered it a minimum standard, and 30% did not express an opinion.
Accessibility	Compliance with accessibility standards should be a minimum requirement in all newly built Ebola and Marburg treatment centres. These standards must be applied across critical areas including waiting zones, screening and triage areas, and wards for suspected and confirmed patients, to ensure equitable access for all individuals, including those with limited mobility.		Ensuring physical accessibility in treatment centres is fundamental to delivering inclusive, dignified, and effective care. Individuals affected by Ebola or Marburg disease may include persons with pre-existing physical disabilities, temporary mobility limitations (e.g., pregnant women or elderly), or patients in critical condition who require stretchers or wheelchairs. When accessibility features such as ramps, wide doorways, and accessible toilets are not integrated into the design, it hinders patient flow, delays care, and burdens caregivers. While constraints in emergency or pre-existing structures may limit full compliance, newly constructed treatment centres offer the opportunity to embed accessibility from the outset. This is not only in line with universal health coverage principles but also improves operational efficiency, benefiting not just patients but also health workers who move patients and equipment. Therefore, accessibility should not be treated as an optional feature but rather as a core standard in the design and construction of future treatment centres. The majority of experts (57%) believe that accessibility should be a minimum requirement, whereas a recommended criterion (32%), while the 11% did not express an opinion.
Waste area	All Ebola and Marburg treatment centres must include a dedicated waste management area. This area should ensure the safe temporary storage, treatment and disposal of infectious waste. Where waste is treated off-site, the design must still provide sufficient space and capacity to handle waste safely until collection and transfer.	The surface area allocated for waste management in Ebola or Marburg treatment centres should be determined based on facility size, number of beds, expected patient load, type and volume of waste generated, and available disposal systems.	Safe waste management is essential for infection control and environmental protection. While there is no fixed formula linking surface area strictly to the number of beds, the space must be sufficient to house key equipment, such as incinerators, ash pits, and shredders, and allow safe workflows. Experts emphasized that excessively large areas are impractical and hard to manage, while undersized zones compromise safety. A range of 2–3 m² per bed provides a practical reference point that balances operational functionality and spatial efficiency. The vast majority of experts (89%) consider the on-site waste area a minimum standard.

Areas and features	Minimum requirement	Recommended feature	Rationale
	In Ebola and Marburg treatment centres, waste disposal sites must be strategically located based on the type of waste and IPC protocols: • Infectious waste, pathological waste, organic waste, and sharps waste should be managed within the high-risk area, as they originate from clinical activities involving infected patients. • Glass and sharps waste from vials, non-hazardous waste, chemical waste should be located in areas straddling both low- and high-risk zones, ensuring both safety and ease of access from different areas. Some flexibility is acceptable depending on the size of the facility and local infrastructure, but waste zones must always follow IPC standards, prevent cross-contamination, and ensure protection for patients, staff, and waste handlers.		Proper zoning of waste disposal areas in Ebola and Marburg treatment centres is essential for effective infection prevention and control. This strategic placement supports operational efficiency, reinforces IPC protocols, and ensures that each type of waste is processed in a manner appropriate to its risk level. Flexibility in design is important, but safety, containment, and regulatory compliance must guide the final layout
Laundry <i>(Reusable PPE and scrubs)</i>	A dedicated on-site laundry service must be available in the low-risk area to reprocess reusable PPE (aprons, boots, goggles) and staff clothing (scrubs). The laundry must include a cleaning and disinfection area, a drying area, and a folding and storage area. Running water and continuous supply of chlorine solution must be available.	The laundry should be strategically located in close proximity to the doffing area, donning area, and staff entrance for efficient workflow. Mechanized washing machines are recommended where feasible. Separate sections with clear physical separation between wet processing and dry areas. Protected storage conditions to prevent dust, moisture, pests, and recontamination.	Items arriving at the laundry have been initially decontaminated at the doffing area. The laundry provides comprehensive cleaning, disinfection, drying, and storage to prepare reusable PPE and staff clothing for safe reuse. On-site laundry with adequate water and chlorine resources supports consistent disinfection standards and reduces occupational exposure risks for laundry and health and care personnel.

Services location

Service	Preferred Location
Ambulance Bay	Straddling high-risk area and outside
Waiting Room	High-risk area
Screening	Straddling low- and high-risk areas
Clinical assessment	High-risk area
Discharge Shower	Straddling high-risk area and outside
Waste Zone	Straddling low- and high-risk areas
Morgue	Straddling high-risk area and outside
Laboratory	Straddling low- and high-risk areas
Survivor Clinic	Outside and nearby the centre
Visitor Area	Straddling high-risk area and outside
Storage for Dirty Items	High-risk area
Laundry	Low-risk area
Reprocessing Equipment Space	Straddling low- and high-risk areas
Storage for Clean Items	Low-risk area
Psychosocial Space – Staff	Low-risk area, Outside and nearby the centre
Psychosocial Space – Family	Outside and nearby the centre
Children and Family Space	Mixed: Low-risk, outside, and straddling zones

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Annexes

Annex 1. Scoping review results

A total of thirteen documents met the predefined inclusion criteria and were carried forward for detailed analysis. These sources included three peer-reviewed journal articles, two of which address Ebola treatment-centre infrastructure and one focusing on Marburg, along with a single conference proceeding. Three documents comprised formally published guidelines, and the remaining six consisted of two technical reports and four letters to the editors.

In the 1995 Kikwit outbreak report (32), Médecins Sans Frontières rapidly rehabilitated an isolation ward at Kikwit General Hospital, restoring both its physical structure and critical utilities. The original 20-bed brick building, comprising the “infected” zone, was supplemented with four tents to establish a clearly demarcated “clean” area for staff and a “neutral” zone for relatives. A chlorinated water-supply system, fed by onsite cisterns and powered by generators, ensured a continuous potable source. Showers, latrines and waste-disposal pits were installed in each zone, while entry and exit points incorporated footbaths filled with hypochlorite solution to enforce unidirectional flow and avoid cross-contamination. This strict zoning, combined with systematic personal protective equipment distribution, correlated with a quick drop in health and care-worker infections, from 67 cases prior to intervention to only three subsequently, and zero new cases among burial teams, demonstrating the decisive impact of rapid infrastructural upgrades on outbreak safety and effectiveness.

The study by Merler and colleagues (15) employs a detailed spatial agent-based model of Liberia’s population to quantify the relative contribution of households, hospitals and funerals to Ebola virus transmission and to evaluate the impact of key non-pharmaceutical interventions, most notably the deployment of Ebola treatment units, implementation of safe-burial protocols and distribution of household protection kits. Drawing on demographic data to place over 3000 grid-cells of synthetic households and calibrating transmission parameters (household, hospital and community) via Markov chain Monte Carlo against reported cases and deaths through mid-August 2014, the analysis estimates that, prior to scale-up of Ebola treatment units, roughly 38% of infections occurred in hospitals, 31% in households and 9% at funerals. Introduction of dedicated Ebola treatment unit beds and progressive uptake of safe burials are credited with halving transmission in clinical settings and driving the epidemic’s downturn after September 2014. Model projections suggest that, by January 2015, the combination of Ebola treatment units and safe-burial measures averted approximately 25 000 cases relative to no-intervention, with additional, but diminishing, gains from household kits at coverage levels above 50%. These findings underscore the critical role of rapid establishment of specialized treatment infrastructure, alongside systemic infection-control measures, in curtailing filovirus spread.

In the 2005 Marburg virus disease response at Uíge Provincial Hospital in Angola, Médecins Sans Frontières integrated an ad-hoc isolation room, where five patients and unremoved corpses had been left untreated, into a purpose-built Marburg ward comprising four separate structures at the hospital’s periphery. Two permanent buildings were dedicated to suspected or probable cases and a third to confirmed cases, with an additional temporary block held in reserve. This arrangement provided clear spatial separation between unconfirmed and confirmed patients, thereby reducing cross-contamination risk. Thick polyethylene fencing enclosed the compound, and strict unidirectional flow was enforced through designated entry and exit points: one for staff donning barrier PPE and another for patients and relatives. Although infection-control procedures successfully prevented any secondary transmission within the ward, the physical separation of “clean” nursing stations from patient areas impeded communication. Post-outbreak reflections recommended relocating staff stations to improve visibility and contact with families, adopting more forgiving construction materials to reduce patient and staff stress, and balancing opaque and transparent fencing to both reassure the community about biohazard containment and demystify isolation practices (32).

You et al. (33) report on the rapid deployment and operational performance of a prefabricated, “E-shaped” Ebola treatment centre constructed near Monrovia. The 5000 m² facility, erected in under 28 days from interlocking infectious-disease-unit panels, was organized into three functional “areas” (contaminated, potentially contaminated and clean), two buffering “zones” and two unidirectional “lines” separating clean and contaminated flows. Sixteen discrete modules housed clinical wards, a patient disinfection unit,

morgue, supply depot and training centre. The buffering lines compelled HCWs to move deliberately when doffing personal protective equipment, ensuring adequate hand- and boot-washing time, while the dual-line supply system allowed uncontaminated materials to enter via a clean pathway and waste to exit exclusively through the contaminated route. Air conditioning throughout the ward improved staff comfort and patient monitoring, supported by a central video-camera system and dual bedside intercoms that enabled real-time vital-sign surveillance and patient-staff communication without breaching containment. No staff infections occurred among 240 HCWs over 180 admissions, and six of ten confirmed cases survived, underscoring the design's effectiveness in preventing cross-contamination and maintaining high standards of care. Early operational experience, however, revealed two key weaknesses. First, airflow and ventilation were inadequate compared with tent-based Ebola treatment centres, leading to chlorine vapour irritation of eyes and airways; a supplementary ventilation system was added to restore comfort. Second, the cost of prefabricated construction remained substantially higher than that of traditional tents, posing challenges for rapid scale-up in resource-limited settings. These lessons highlight both the strengths of modular, zone-based architectures for infection control and the need to optimize air-handling and cost efficiencies in future centre designs.

You et al. (33) describe the rapid construction and operational performance of an “E-shaped” Ebola treatment centre in Liberia, built entirely from movable prefabricated panels to meet infectious-disease-unit standards. The facility enforces strict unidirectional flows by dividing its footprint into three sequential zones, cold (clean), warm (buffer) and hot (contaminated), linked by two one-way corridors. Clinical staff don basic PPE (scrub suit, N95/N99 mask, goggles, gown, inner gloves, boots) in the cold zone, proceed through the buffer where advanced PPE (waterproof apron, face shield, outer gloves, shoe covers) is added, and enter the patient wards only after passing through hypochlorite footbaths. Between January and June 2015, the centre admitted 180 patients, ten of whom were laboratory-confirmed Ebola cases; six survived. Crucially, no clinical staff infections were recorded. The authors conclude that this zone-based, prefabricated architecture, by embedding PPE workflows into the very layout of the ward, enables rapid deployment, robust infection control and effective patient care without compromising healthcare-worker safety.

Devi et al. (34) introduce the ALIMA “CUBE”, a transparent, negative-pressure chamber deployable in under 48 hours at an approximate cost of € 16 000. Glove-port interfaces permit bedside care without full PPE changes, reducing exposure risk and workload while preserving family contact and patient morale. Deployment of twelve units in North Kivu achieved enhanced staff safety and patient acceptance, suggesting that modular biocontainment chambers can humanize Ebola care while maintaining rigorous infection control.

A conference proceeding (35) describes how rigorous WASH measures are embedded within a three-zone layout to safeguard both patients and staff. In this model, “very low-risk” areas, offices, warehouse, pharmacy and kitchen, are physically separated from “low-risk” support zones such as laundry and coordination hubs, which in turn are segregated from the “high-risk” patient-care sector encompassing waste management and the morgue. Access to each zone is tightly controlled by a system of gates, permitting only essential personnel to transit between areas. Within the high-risk sector, further subdivision ensures granularity of containment: Zone 1 comprises the ambulance bay, triage and suspect-case ward; Zone 2 accommodates probable or high-suspect patients; and Zone 3 houses confirmed cases alongside a recovery ward, dedicated waste-disposal area, morgue and morgue ambulance bay. This multilayered spatial separation is presented as critical to preventing cross-infection between non-EVD and EVD patients and maintaining staff safety through integrated WASH and IPC protocols.

The Médecins Sans Frontières Operational Centre Brussels commissioned a comprehensive review (36) of thirteen Ebola treatment centres established by Médecins Sans Frontières between March 2014 and March 2015, spanning bed capacities from twenty up to 250. This review, conducted as part of the Operational Centre Brussels Ebola evaluation, identified five distinct phases in the evolution of Ebola treatment centre design and construction as response needs grew and shifted. Although the earliest centres adhered to WHO's 2008 guidelines, rapid increases in patient load and the sustained nature of the epidemic necessitated multiple rounds of design revision, lessons that were often shared orally rather than formally documented. Training programmes in Brussels introduced frontline staff to core concepts such as high-risk and low-risk zoning, but did not address the full spectrum of critical design elements, leading to gaps in operational readiness. Analysis revealed that larger and more complex centres require advanced design tools (for example, computer-aided

design software) and a high degree of on-site coordination, ideally with the design team embedded within the construction crew. Collaborative construction efforts frequently encountered delays when external stakeholders were not fully aligned on timelines or site requirements. Moreover, insufficient integration of site-specific characteristics, urban versus rural conditions, soil drainage, local materials, into initial design reflections often produced operational challenges when centres were adapted or expanded. The review further emphasized the necessity of a holistic design overview that maps all Ebola treatment centre functions, physical structure, space allocation, water and waste flows, laboratory operations, supply chains, staff support areas and even non- Ebola treatment centre human activities such as family visiting zones, into a single, coherent blueprint. Such complexity cannot be conveyed adequately through simple sketches or verbal instruction alone but demands appropriate digital design tools. Finally, the availability and sourcing of construction materials must be factored into the design process from the outset, particularly for facilities intended to operate beyond twelve months. This evaluation underscored the importance of rigorous, documented design frameworks and adaptive planning to ensure that centres remain safe, functional and resilient throughout protracted outbreak responses.

The *INITIATE*²'s 2023 *Technical Report on the Infectious Disease Treatment Module* (37) synthesizes decades of field experience, engineering innovation and infection-control science into a single, deployable blueprint. Drawing on lessons from filovirus responses, this document identifies forty-nine interdependent design elements that collectively ensure rapid assembly, robust biocontainment, reliable resource management and humane patient care under the most austere conditions. Organized around structural modularity, infection prevention and control, resource sustainability and human-centred inclusivity, these components define the minimum specifications and aspirational features for any facility, or “pod”, intended to be trucked, air-lifted or carried by hand and operational within forty-eight hours of arrival.

- **Structural modularity and transportability.** The module must be built from interchangeable pods that connect via a simple clamp-and-seal system, deployable in minutes by unskilled teams and sized for transport in pick-up trucks or even by hand. A smooth, hard-surface flooring system snaps together atop a reinforced subframe, while an outer pneumatic envelope provides wind-resistance up to 80 km/h. Large, identical openings on every façade permit attachment of transparent insert panels, complete with glove-port apertures and material-transfer hatches, enabling sliding furniture, medical-device alcoves and family porches to be reconfigured as needed.
- **Infection prevention and control.** A strict three-zone layout is enforced by integrated shutters, hypochlorite footbaths and one-way gating. Controlled building ventilation is achieved through screened inlet windows, natural-ventilation louvres sized to deliver ≥ 160 L/s when open, and provisions for simple mechanical exhaust or air-conditioning add-ons. Water- and wastewater-management panels allow pour-flush latrines, shower drains and sludge outlets to be piped directly into a closed collection network. Materials throughout are chosen for rapid disinfection, PVC-covered floors, sealed transparent surfaces, zero-leakage glove inserts and vector-proof mosquito nets.

Resource-management and sustenance. Each two-bed module includes plug-in collars for dual water inlets, bladder-fed storage and booster-pump connections, alongside built-in solar-PV readiness and modular fan-coil units for heating or cooling. Durable, repairable parts minimize local maintenance needs, while “cradle-to-cradle” design ensures packaging, membranes and structural tubes may be repurposed or recycled at end-of-life.

- **Human-centred care and inclusivity.** A central “green” staff pod affords continuous observation without donning full PPE, while two individual “red” patient rooms may each accommodate two occupants in emergencies. Transparent screens on both bedhead and bedside walls maximize sightlines, and integrated intercoms support remote communication. Family-viewing porches in the morgue, accessible latrine/shower stalls sized for wheelchair users, clear wayfinding and ergonomic glove-port positioning (covering 5th–95th percentile anthropometry) together uphold dignity, privacy and cultural adaptability.

The Centers for Disease Control and Prevention 2024 *Interim Guidance for Preparing Ebola Treatment Centres* (38) establishes a comprehensive framework for United States hospitals designated to receive or manage patients with suspected or confirmed Ebola virus disease. It mandates clear spatial zoning, from

screening and triage through dedicated patient wards, and requires engineering controls such as negative-pressure rooms or dedicated HVAC systems, hands-free handwashing stations and seamless, easily decontaminable surfaces. The document integrates donning and doffing protocols into the facility layout, prescribes waste-stream separation and environmental monitoring, and specifies staffing models and training requirements to ensure that structural features and operational procedures together protect HCWs and prevent environmental contamination.

Oxfam's 2015 guidance on *Medical Isolation Facilities in Outbreak Responses* (39) addresses the challenges of low-resource settings by prescribing a modular, three-zone architecture comprising clean support areas, buffer ("grey") zones and high-risk patient wards. Colour-coded signage and simple physical barriers enforce unidirectional footpaths, while essential WASH infrastructure, including chlorinated water supply, pour-flush latrines adjacent to patient areas and secure waste pits, ensures basic hygiene standards. The manual also provides site selection criteria (drainage, accessibility and proximity to community resources), recommendations for locally sourced modular shelter materials, and strategies for community engagement to align facility design with cultural practices and enhance local acceptance.

Médecins Sans Frontières' 2008 *Filovirus Haemorrhagic Fever Guideline* (40) synthesizes field experience from multiple Ebola and Marburg disease outbreaks into a unified, rapid-deployment blueprint. Core infrastructure elements include strictly demarcated patient and staff zones linked by anterooms, unidirectional flow enforced by hypochlorite footbaths and gate controls, and dedicated water-and-waste systems with closed-loop drainage and on-site disinfection. The guideline specifies room dimensions, ventilation requirements and PPE transition workflows, and embeds triage algorithms, staffing ratios and waste-handling protocols within the architectural design.

Annex 2. Stakeholders' consultation results

System-level topics: More than one-third of respondents (37%) indicated that all proposed topics should be addressed. When considered individually, the most frequently endorsed topics were location requirements for site selection (34%), structural requirements (materials, durability) (32%), water-supply quality and quantity (32%), and size/bed-capacity standards (29%). Electricity requirements, excreta (toilet) management and accessibility/inclusiveness each garnered support from 29% of stakeholders. Emerging priorities voiced in free-text comments included gender-sensitive and child-friendly design features, post-discharge social and economic reintegration support, and the integration of immunization, expanded laboratory functions, intensive-care and delivery units.

Services and functional areas: Nearly half of all participants (45%) selected every proposed service and area as essential. Among discrete items, patient screening/triage (29%), confirmed-patient wards (29%), and suspect-patient wards (28%), were most commonly prioritized. Waste-disposal zones and ambulance bays each received endorsement from 28% and 25% of respondents, respectively. Donning and doffing areas, morgue facilities, staff and patient sanitation, laboratories and convalescent (survivors') wards were similarly highlighted. Free-text responses further recommended dedicated survivors' clinics, children's play and family-meeting areas, psychological-support rooms and specialized intensive care unit level treatment spaces.

Interpretation: The strong support for "all of the above" across both topic and service lists underscores a clear mandate for comprehensive guidance that spans site selection, infrastructure design, operational workflows and patient-centred care. Core technical priorities, such as precise zoning, utility provision, waste management and bed-capacity norms, emerged alongside equity-focused considerations like accessibility, gender-sensitivity and post-discharge reintegration. These findings will directly shape the evidence-to-decision framework, ensuring that the final normative standard addresses both the fundamental engineering requirements and the broader human-centred dimensions of filovirus treatment-centre design.

Annex 3. Assessment tool

Section 1: Health care facility identification and assessment information

1.1. Assessment information			
No.	Question	Response options	Comment
1.1.1	Date of assessment (DD/MM/YYYY)		
1.1.2	Name of the person conducting the assessment		
1.1.3	email		<i>Of the person conducting the assessment</i>
1.1.4	phone		<i>Of the person conducting the assessment</i>
1.2. Geographical location			
No.	Question	Response options	Comment
1.2.1	Country		
1.2.2	Region/province name		
1.2.3	District name		
1.2.4	City / village / clan / locality name		
1.2.5	Address of facility / GPS coordinates		
1.2.6	Record of facility location	1. Urban 2. Peri-/ ex-urban 3. Rural	
1.3. Facility identification			
No.	Question	Response options	Comment
1.3.1	Facility code		
1.3.2	Facility name		

No.	Question	Response options	Comment
1.3.3	Managing authority	1. Government / public 2. Private for profit 3. Private not for profit (e.g., nongovernmental organization, faith-based) 4. Military / police / national guard 5. University 6. Other	
1.3.4	Facility director/manager's name		
1.3.5	Facility director/manager's telephone number		
1.3.6	Facility director/manager's email address		
1.3.7	Respondent or key informant's name		
1.3.8	Respondent or key informant's position		
1.3.9	Type of structure	1. Permanente facility (existing facility built with i.e. concrete, bricks, etc.) 2. Temporary (medium-term) facility (i.e. concrete slabs and wooden structure, rub halls, etc.) 3. Temporary (short-term) facility (i.e., tents, etc.)	
1.3.10	Type of facility	1. Screening and temporary isolation (i.e., primary health clinic with screening and temporary isolation capacity) 2. Ebola/Marburg stand-alone treatment centre	<i>If answer "1" → go to the next question 1.3.11</i> <i>If answer "2" → skip the next questions, go to "section 2" + "section 4"</i>
1.3.11	Type of existing health care facility	1. First referral hospital / district hospital 2. Regional / provincial referral hospital 3. National referral hospital 4. Other general hospital with specialties or single-specialty hospital 5. Infectious diseases unit 6. Primary health clinic 7. Other	

1.4 Existing health care facility services

No.	Question	Response options	Comment
1.4.1	The facility offers outpatient services	1. Yes 2. No	
1.4.2	The facility offers inpatient services	1. Yes 2. No	If answer “no” → skip the next questions and go to “section 2” + “section 3”
1.4.3	Total number of overnight/ inpatient beds	 beds	
1.4.4	Of the total number of inpatient beds, how many are intensive care unit beds?	 beds	This number indicated in this answer should be minor than the number indicated in question 1.4.3
1.4.5	The facility has dedicated 24-hour staffed emergency unit	1. Yes 2. No	
1.4.6	The facility has intensive care or high-dependency unit	1. Yes 2. No	

Section 2: Overall facility assessment

This section is applicable to all the health facilities addressed in this document. It is a preliminary and generic assessment of the overall facility. For every area, select one of the options meets target, partially meets target, does not meet target, do not know (DK), and not applicable (NA).

2.1 Facility accesses and flows

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
2.1.1	Health care facility entrances	Patients and staff have dedicated entrances clearly labelled.	There are two different entrances for patients and staff but NOT properly labelled or NOT properly used.	There is only one common entrance for both staff and patients.					
2.1.2	Area distribution	Two areas are clearly divided and identified: one for staff in PPE and patients (high-risk zone), AND one for staff only (low-risk zone).	Two areas are divided: one for staff in PPE and patients (high-risk zone), AND one for staff only (low-risk zone) but NOT clearly identified.	There is NO clear division between staff-only (low-risk) area and staff in PPE and patients' area (high-risk).					
2.1.3	Patient flow	The patient flow is logical, clear, properly labelled, and easy to follow. The spaces are allocated to support a unidirectional flow: waiting room > screening and triage> treatment wards (suspect-confirmed) or temporary isolation room > discharge or morgue.	The patient flow is logical, but NOT clear, NOT labelled, and NOT easy to follow. The spaces are allocated to support a unidirectional flow: waiting room > screening and triage> treatment wards (suspect-confirmed) or temporary isolation room > discharge or morgue.	The patient flow is NOT logical, NOT clear, NOT labelled, and NOT easy to follow.					
2.1.4	Staff flow	All staff access the facility through staff entrance > syndromic surveillance (questionnaire/assessment) > changing room > staff area > donning > high-risk zone > doffing > staff area; the flow is rational, clear and properly labelled.	All staff access the facility through staff entrance > syndromic surveillance (questionnaire/assessment) > changing room > staff area > donning > high-risk zone > doffing > staff area; the flow is rational and clear but NOT labelled.	There is NOT a proper flow for staff.					
2.1.5	Visitor area	A dedicated visitor area outside the patients' ward is available. A double fence ensures physical separation between visitor area and treatment ward. Visitors have safe access to all patients.	A dedicated visitor area outside the patients' ward is available. A double fence ensures physical separation between visitor area and treatment ward. Visitors have safe access to some patients.	There is NO visitor area and no protocols to ensure visitors access to patients.					

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
2.1.6	Facility delimitation	A double fence is delimitating the compound of the facility. Few accesses are available and all the accesses are controlled.	A double fence is delimitating the compound of the facility. Few accesses are available, but NOT all the accesses are controlled.	The facility is not delimited by a continuous double fence; anybody can access the compound.					

2.2 Facility systems and finishing

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
2.2.1	Electricity	The electrical system is compliant with national regulations and equipped with a back-up energy supply system.	The electrical system is compliant with national regulations, but critical services are NOT connected to the back-up power supply system.	The electrical system does not meet national regulations OR a back-up power supply system is not available.					
2.2.2	Fire safety	The facility's fire system is compliant with national regulations, and an evacuation plan is available and known by staff and patients.	The facility's fire system is compliant with national regulations, but the evacuation plan is NOT available.	The facility's fire system is not compliant with national regulations.					
2.2.3	Ventilation, airflow rate	All the areas have a minimum indoor ventilation with the presence of at least one operable window with a window-to-floor ratio of at least 1:8.	All the areas are well ventilated with the presence of at least one windows always open; however, the window-to-floor ration is less than 1:8.	All the areas are not equipped with windows OR windows are closed most of the time.					
2.2.4	Ventilation system maintenance (if mechanical)	The mechanical ventilation system is perfectly functioning and properly maintained (compliant with the manufacturer recommendations).	The mechanical ventilation system is perfectly functioning, but maintenance is not regular as per manufacturer recommendations.	The mechanical ventilation system is not functioning OR maintenance has never been done.					
2.2.5	Finishing materials and furniture	The building finishing materials and furniture are smooth, non-porous, easy to maintain and repair, and resistant to microbial growth.	The building finishing materials and furniture are smooth, non-porous and resistant to microbial growth but NOT easy to maintain and repair.	The building finishing materials and furniture are NOT smooth, non-porous or resistant to microbial growth.					

2.3 Waste management

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
2.3.1	Waste area	A dedicated and fenced waste area is available, of sufficient capacity to treat and manage ash, organic and sharp waste; a cleaning and disinfection area with running water is available in the waste storage area.	A dedicated waste area is available but not fenced or secure or not sufficient capacity or all waste grouped together, or a cleaning and disinfection area with running water is not available.	There is NO dedicated waste storage area available.					
2.3.2	Waste segregation	A functional 3-bin waste system for segregating waste exists and waste is correctly segregated at all waste generation points.	A functional 3-bin waste system exists at some points, but not all waste generation points or waste is not correctly segregated.	No waste bins exist.					
2.3.3	Hazardous waste	Hazardous waste is safely treated or disposed of on-site or off-site.	On site-treatment technology is not of sufficient capacity or waste is not collected regularly for off-site treatment.	Hazardous waste is not safely treated or disposed of, and added to general waste.					

2.4 Water and sanitation

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
2.4.1	Water availability	Water is available on premises from an improved source and of sufficient quantity.	Water is available from an improved source but NOT on premises or of sufficient quantity.	Water is not available, or available from an unimproved source.					
2.4.2	Water storage	Water storage is available and protected and sufficient for at least two days' needs.	Water storage is available but not protected or not sufficient for two days.	Water storage is not available.					

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
2.4.3	Water quality	Drinking-water quality is compliant with national regulation; in emergency settings, water has appropriate chlorine residual (FRC 0.2–0.5 mg/L at point of use) or 0 <i>E. coli</i> /100 mL and turbidity <5 NTU (nephelometric turbidity units).	Drinking-water quality is compliant with national regulation; in emergency settings, water has appropriate chlorine residual (FRC 0.2–0.5 mg/L at point of use) or 0 <i>E. coli</i> /100 mL and turbidity <5 NTU.	Drinking-water quality is NOT compliant with national regulation; water has less than 0.2 mg/L chlorine residual at point of use or presence of <i>E. coli</i> or turbidity >5 NTU.					
2.4.4	Wastewater	Wastewater is safely managed by preventing the discharge of wastewater and faecal sludge to the environment.	Wastewater is partially safely managed, some leakages exist.	Wastewater is discharged into the environment.					
2.4.5	Hand washing	All toilets have functional hand washing stations within 5 metres.	At least 50% of toilets have functioning hand washing stations within 5 metres.	Fewer than 50% of toilets have functioning hand washing stations within 5 metres.					
2.4.6	Improved toilet for menstrual hygiene management (only in case confirmed patients are cohorted)	At least one usable improved toilet meets menstrual hygiene management needs.	Menstrual hygiene management toilet is not available, or menstrual hygiene management toilet is available but not usable OR in the menstrual hygiene management toilet there is space for women to wash but no water available, toilet is not clean/ in disrepair or bin for disposal of waste is available but full.	Menstrual hygiene management facility is not available, or a facility is available, but toilet is not usable.					
2.4.7	Toilet for people with reduced mobility	At least one improved usable toilet meets the needs of people with reduced mobility.	Toilet meets needs of people with reduced mobility but is not usable or toilet is usable but only partially meets needs of people with reduced mobility. No toilets for people with reduced mobility or toilets are not usable.	No toilets for people with reduced mobility or toilets are not usable.					

Section 3: Screening with temporary isolation

This section is applicable to primary health care and other possible points of entry to the health system without treatment capacity. For every area, select one of the options meets target, partially meets target, does not meet target, do not know (DK), and not applicable (NA).

3.1 Waiting room

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
3.1.1	Location	The waiting room is located in proximity (less than 5 metres) to the screening and triage areas.	The waiting room is located between 5 metres and 15 metres to the screening and triage areas, but the path is easy to follow.	The waiting room is located at more than 15 metres to the screening and triage areas, and the path is not easy to follow.					
3.1.2	Physical distance and furniture	In the waiting room, the furniture (chairs, fences, ground marks) ensures there is at least 2 metres between patients (1 metre for IPC + 1 metre for health worker movement).	In the waiting room, the furniture (chairs, fences, ground marks) ensures there is at least 1 metre between patients.	In the waiting room, the furniture (chairs, fences, ground marks) is randomly located or not present therefore there is LESS than 1 metre between patients.					
3.1.3	Floor surface	There is at least 2.5 m ² per patient.	There is between 2.5 m ² and 1.9 m ² per patient OR there are individual booths (physical separation between patients).	There are NO individual booths and there is LESS than 1.9 m ² per patient.					
3.1.4	Surge plan	The waiting area can be extended in case of the patients accesses increase, avoiding overcrowding. It is already equipped and ready to use.	The waiting area can be extended in case of the patients accesses increase, avoiding overcrowding. However, it is NOT equipped.	There is NO possibility to extend the waiting room.					
3.1.5	Toilet facility	There are dedicated toilets directly accessible from the waiting area, that fully meet the criteria listed in section 2.4.5.	There are dedicated toilets NOT directly accessible from the waiting area OR that fully meet the criteria listed in section 2.4.5.	There is no dedicated toilet OR there is dedicated toilet that do not meet the criteria listed in section 2.4.5.					

3.2 Screening and clinical assessment

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
3.2.1	Location	A screening area is placed at the entrance of the facility. All patients pass through the screening before accessing the facility. The screening is visible, clearly labelled and proper wayfinding guides the flow of patients.	A screening area is placed at the entrance of the facility. All patients pass through the screening before accessing the facility. However, it is not clearly labelled AND no dedicate wayfinding is in place to guide the flow of patients.	NOT all patients pass by the screening before accessing the facility.					
3.2.2	Patient's pathway	In the screening area, there are separated exits for patient who meet the case definition (suspect cases) and patient who does not meet the case the definition (non-suspect cases) to ensure separate patients' flows. Suspect cases access the screening facility while non-suspect cases proceed with their path.	In the screening area, there is only one exit for patient who meet the case definition (suspect cases) and patient who does not meet the case the definition (non-suspect cases), however the shared path is short, and suspect cases access the screening facility while non-suspect cases proceed with their path.	In the screening area, there is only one exit for patient who meet the case definition (suspect cases) and patient who does not meet the case the definition (non-suspect cases). Non-suspect cases may access by accident the screening facility rather proceeding with their path.					
3.2.3	Physical distance and furniture	There is at least 2 metres distance, between patients and staff when conducting screening. Physical barrier (fence or furniture) or marks on the ground support keeping the distance.	There is LESS than 2 metres distance between patients and staff OR no marks on the ground or physical barriers helps in keeping the physical distance.	There is NO distance between patients and staff.					
3.2.4	Floor surface	For each screening station there is at least 12 m ² of surface available. In settings where clinical assessment (involving close physical contact and PPE use) is involved, additional 6 m ² of surface is available.	For each screening station, there is between 12 m ² and 8 m ² of surface available. In settings where clinical assessment (involving close physical contact and PPE use) is involved, additional surface between 6 m ² and 4 m ² is available.	For each screening and triage station, there is less than 4 m ² of surface available. In settings where clinical assessment (involving close physical contact and PPE use) is involved, additional surface is less than 4 m ² .					

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
3.2.5	Donning/ doffing (putting on/ taking off PPE) location	There are dedicated donning and doffing spaces for the screening station that are strategically located between the low-risk zone and the high-risk zone.	There are NO designated spaces for donning and doffing dedicated to the screening, but staff can use donning and doffing spaces, that are strategically located between the low-risk zone and the high-risk zone, and that are in proximity to the screening a station (less than 30 metres).	There are NO designated spaces for donning and doffing dedicated to the screening and staff put on and take off the PPE in donning and doffing rooms far from screening (more than 30 metres) OR donning and doffing are not strategically located between the low-risk zone and the high-risk zone.					
3.2.6	Donning dimensions and furniture	The donning area is big enough to ensure space for shelves for PPE storage, bins, mirror, bench/chair, small table. The dimension of the donning and doffing areas also ensures optimal free movement of 1 staff (at least 16 m ²).	The donning area is big enough to ensure space for shelves for PPE storage, bins, mirror, a chair. The dimension of the donning and doffing areas also ensures free movement of 1 staff (between 16 m ² and 9 m ²).	The donning is not big enough to ensure space for basic furniture and PPE storage AND staff has no sufficient space to properly perform donning procedure (less than 9 m ²).					
3.2.7	Doffing dimensions and furniture	The doffing area is big enough to ensure space for full rotation with open arms for 1 staff, 2 metres distance with the second staff monitoring, bins and handwashing station. The doffing area (1 line) is at least 12 m ² .	The doffing area is big enough to ensure 2 m distance with the second staff monitoring, bins and handwashing station. The full rotation with open arms of staff taking off the PPE is difficult. The doffing area (1 line) is between 12 m ² and 10 m ²	The doffing area is not big enough to ensure full movement of the staff OR to ensure 2 metres distance between the 2 staff OR to host bins and handwashing station. The doffing area (1 line) is less than 10 m ² .					
3.2.8	Discharge shower location	A discharge shower is available, and every non-case is passing through it before exit the screening area. It is located in proximity (less than 10 metres) to the screening areas.	A discharge shower is available, and every non-case is passing through it before exit the facility. It is located far (more than 10 metres) from the screening areas.	A discharge shower is not available OR it is available but not every non-case pass through it before exit the facility.					
3.2.9	Discharge shower dimensions and furniture	The discharge shower is at least 4 m ² . There is space for dressing, undressing, a shower and to ensure the flow is unidirectional.	The discharge shower is between 3 m ² and 4 m ² . There is space for dressing/undressing, and a shower.	The discharge shower is less than 3 m ² . Shower and changing zone are sharing the same area.					

3.3 Temporary isolation rooms

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
3.3.1	Location and access	The isolation room is a single room, clearly identified and labelled, and access is controlled. A dedicated pathway leads suspect patients from the screening to the isolation room without walking into the staff area. Individual toilets are available. Suspect cases wait here for the referral to the suitable health facility, a dedicated exit toward the ambulance bay is available.	The isolation room is a single room, clearly identified and labelled, and access is controlled. A dedicated pathway leads suspect patients from the screening to the isolation room without walking into the staff area. Individual toilets are available. Suspect cases wait here for the referral to the suitable health facility, but there is NO dedicated exit toward the ambulance bay.	The isolation room is NOT a single room OR NOT clearly identified and labelled OR access is NOT controlled. NO dedicated pathway leads suspect patients from the screening to the isolation room without walking into the staff area, NO dedicated exit toward the ambulance bay is available.					
3.3.2	Donning/ doffing (putting on/ taking off PPE) location	There are dedicated donning and doffing spaces for the isolation room that are strategically located between the low-risk zone and the high-risk zone.	There are NO designated spaces for donning and doffing dedicated to the isolation room, but staff can use donning and doffing spaces, that are strategically located between the low-risk zone and the high-risk zone, and that are in proximity to the isolation room (less than 30 metres).	There are NO designated spaces for donning and doffing dedicated to the isolation room and staff put on and take off the PPE in donning and doffing rooms far the isolation room (more than 30 metres) OR donning and doffing are not strategically located between the low-risk zone and the high-risk zone.					
3.3.3	Donning dimensions and furniture	The donning area is big enough to ensure space for shelves for PPE storage, bins, mirror, bench/chair, small table. The dimension of the donning and doffing areas also ensures optimal free movement of 1 staff (at least 16 m ²).	The donning area is big enough to ensure space for shelves for PPE storage, bins, mirror, a chair. The dimension of the donning and doffing areas also ensures free movement of 1 staff (between 16 m ² and 9 m ²).	The donning is not big enough to ensure space for basic furniture and PPE storage AND staff has no sufficient space to properly perform donning procedure (less than 9 m ²).					

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
3.3.4	Doffing dimensions and furniture	The doffing area is big enough to ensure space for full rotation with open arms for 1 staff, 2 metres distance with the second staff monitoring, bins and handwashing station. The doffing area (1 line) is at least 12 m ² .	The doffing area is big enough to ensure 2 m distance with the second staff monitoring, bins and handwashing station. The full rotation with open arms of staff taking off the PPE is difficult. The doffing area (1 line) is between 12 m ² and 10 m ² .	The doffing area is not big enough to ensure full movement of the staff OR to ensure 2 metres distance between the 2 staff OR to host bins and handwashing station. The doffing area (1 line) is less than 10 m ² .					
3.3.5	Patients' placement	Suspect cases are in individual rooms with private toilets. (further criteria for toilets in section 2.4.5).	Suspect cases are in individual room with shared toilets. (further criteria for toilets in section 2.4.5).	No individual room for suspect cases is available OR Shared toilets are not cleaned after each use. (further criteria for toilets in section 2.4.5).					
3.3.6	Floor surface	The individual room for suspect cases is at least 12 m ² . Private toilet is available.	The surface of the individual room for suspect cases is between 12 m ² and 9 m ² .	The isolation room is less than 9 m ² .					
3.3.7	Surge plan	Extra rooms are available in case the beds demand increase; it is equipped and ready at any time.	Extra rooms are available in case the beds demand increase, but it is NOT equipped.	There is NO extra rooms are available in case the beds demand increase.					

Section 4: Stand-alone treatment centre

For every area, select one of the options meets target, partially meets target, does not meet target, do not know (DK), and not applicable (NA).

4.1 Waiting room

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
4.1.1	Location	The waiting room is located in proximity (less than 5 metres) to the screening area.	The waiting room is located between 5 metres and 15 metres to the screening area, but the path is easy to follow.	The waiting room is located at more than 15 metres to the screening and area, and the path is not easy to follow.					
4.1.2	Physical distance and furniture	In the waiting room the furniture (chairs, fences, ground marks) ensures there is at least 2 metres between patients (1 metre for IPC + 1 metre for health worker movement).	In the waiting room the furniture (chairs, fences, ground marks) ensures there is at least 1 metre between patients.	In the waiting room the furniture (chairs, fences, ground marks) is randomly located or not present therefore there is LESS than 1 metre between patients.					
4.1.3	Floor surface	There is at least 2.5 m ² per patient.	There is between 2.5m ² and 1.9m ² per patient OR there are individual booths (physical separation between patients).	There are NO individual booths and there is LESS than 1.9 m ² per patient.					
4.1.4	Surge plan	The waiting area can be extended in case of the patients accesses increase, avoiding overcrowding. It is already equipped and ready to use.	The waiting area can be extended in case of the patients accesses increase, avoiding overcrowding. However, it is NOT equipped.	There is NO possibility to extend the waiting room.					
4.1.5	Toilet facility	There are dedicated toilets directly accessible from the waiting area, that fully meet the criteria listed in section 2.4.	There are dedicated toilets NOT directly accessible from the waiting area OR that fully meet the criteria listed in section 2.4.	There are no dedicated toilets OR there are dedicated toilets that do not meet the criteria listed in section 2.4.					

4.2 Screening and Clinical assessment

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
4.2.1	Location	A screening area is placed at the entrance of the facility. All patients pass through the screening before accessing the facility. The screening is visible, clearly labelled and proper wayfinding guides the flow of patients.	A screening area is placed at the entrance of the facility. All patients pass through the screening before accessing the facility. However, it is not clearly labelled AND no dedicate wayfinding is in place to guide the flow of patients.	NOT all patients pass by the screening before accessing the facility.					
4.2.2	Patient's pathway	In the screening area, there are separated exits for patient who meet the case definition (suspect cases) and patient who does not meet the case the definition (non-suspect cases) to ensure separate patients' flows. Suspect cases access the screening facility while non-suspect cases proceed with their path.	In the screening area, there is only one exit for patient who meet the case definition (suspect cases) and patient who does not meet the case the definition (non-suspect cases), however the shared path is short, and suspect cases access the screening facility while non-suspect cases proceed with their path.	In the screening area, there is only one exit for patient who meet the case definition (suspect cases) and patient who does not meet the case the definition (non-suspect cases). Non-suspect cases may access by accident the screening facility rather proceeding with their path.					
4.2.3	Physical distance and furniture	There is at least 2 metres distance, between patients and staff when conducting screening. Physical barrier (fence or furniture) or marks on the ground support keeping the distance.	There is LESS than 2 metres distance between patients and staff OR no marks on the ground or physical barriers helps in keeping the physical distance.	There is NO distance between patients and staff.					
4.2.4	Floor surface	For each screening station there is at least 12 m ² of surface available. In settings where clinical assessment (involving close physical contact and PPE use) is involved, additional 6 m ² of surface is available.	For each screening station, there is between 12 m ² and 8 m ² of surface available. In settings where clinical assessment (involving close physical contact and PPE use) is involved, additional surface between 6 m ² and 4 m ² is available.	For each screening and triage station, there is less than 4 m ² of surface available. In settings where clinical assessment (involving close physical contact and PPE use) is involved, additional surface is less than 4 m ² .					

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
4.2.5	Donning/ doffing (putting on/ taking off PPE) location	There are dedicated donning and doffing spaces for the screening station that are strategically located between the low-risk zone and the high-risk zone.	There are NO designated spaces for donning and doffing dedicated to the screening, but staff can use donning and doffing spaces, that are strategically located between the low-risk zone and the high-risk zone, and that are in proximity to the screening station (less than 30 metres).	There are NO designated spaces for donning and doffing dedicated to the screening and staff put on and take off the PPE in donning and doffing rooms far from screening (more than 30 metres) OR donning and doffing are not strategically located between the low-risk zone and the high-risk zone.					
4.2.6	Donning dimensions and furniture	The donning area is big enough to ensure space for shelves for PPE storage, bins, mirror, bench/chair, small table. The dimension of the donning and doffing areas also ensures optimal free movement of 1 staff (at least 16 m ²).	The donning area is big enough to ensure space for shelves for PPE storage, bins, mirror, a chair. The dimension of the donning and doffing areas also ensures free movement of 1 staff (between 16 m ² and 9 m ²).	The donning is not big enough to ensure space for basic furniture and PPE storage AND staff has no sufficient space to properly perform donning procedure (less than 9 m ²).					
4.2.7	Doffing dimensions and furniture	The doffing area is big enough to ensure space for full rotation with open arms for 1 staff, 2 metres distance with the second staff monitoring, bins and handwashing station. The doffing area (1 line) is at least 12 m ² .	The doffing area is big enough to ensure 2 metres distance with the second staff monitoring, bins and handwashing station. The full rotation with open arms of staff taking off the PPE is difficult The doffing area (1 line) is between 12 m ² and 10 m ² .	The doffing area is not big enough to ensure full movement of the staff OR to ensure 2 metres distance between the 2 staff OR to host bins and handwashing station. The doffing area (1 line) is less than 10 m ² .					
4.2.8	Discharge location	A discharge shower is available, and every non-case is passing through it before exit the screening area. It is located in proximity (less than 10 metres) to the screening area.	A discharge shower is available, and every non-case is passing through it before exit the facility. It is located far (more than 10 metres) from the screening area.	A discharge shower is not available OR it is available but not every non-case pass through it before exit the facility.					

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
4.2.9	Discharge dimensions and furniture	The discharge shower is at least 4m ² . There is space for dressing, undressing, a shower and to ensure the flow is unidirectional	The discharge shower is between 3 m ² and 4 m ² . There is space for dressing/undressing, and a shower.	The discharge shower is less than 3 m ² . Shower and changing zone are sharing the same area.					

4.3 Treatment wards

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
4.3.1	Patient's pathway	The ward is clearly identified and labelled and access is controlled; A dedicated pathway lead suspect or confirmed patients from the screening to the treatment ward without walking into the staff area; discharged patients (tested negative or referred to other facilities) have a dedicated discharge pathway.	The ward is clearly identified and labelled, and access is controlled; A dedicated pathway lead suspect or confirmed patients from the screening to the treatment ward without walking into the staff area; there is NOT a dedicated discharge pathway for discharged patients (tested negative or referred to other facilities).	The ward is NOT clearly identified and labelled OR access is NOT controlled; NO dedicated pathway lead suspect or confirmed patients from the screening to the treatment ward without walking into the staff area.					
4.3.2	Area distribution	There is a dedicated area for staff only (low-risk), with visibility of the patients' rooms	There is a dedicated area for staff only but there is NO visibility of the patients' rooms.	There is NOT a dedicated area for staff.					
4.3.3	Patients' placement: suspect and confirmed wards	There are separated wards for suspects and confirmed cases.	There are no separated wards for suspect and confirmed cases, but all individuals' rooms are available.	There are no separated rooms for suspect and confirmed cases and not all the rooms are individual.					
4.3.4	Suspect cases ward	Suspect cases are in individual rooms with private toilets.	Suspect cases are in individual room with shared toilets.	No individual room for suspect cases is available OR Shared toilets are not cleaned after each use OR suspect and confirmed patients shared the same space.					

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
4.3.5	Confirmed cases ward	Confirmed cases are in individual rooms OR are cohorted in shared rooms with at least 2 metres between patients (1 metre for IPC + 1 metre for health worker movement).	Confirmed cases are cohorted in shared rooms with 1 metre between patients.	Confirmed cases are cohorted in shared rooms with less than 1 metre between confirmed patients OR suspect and confirmed patients share the same space.					
4.3.6	Special needs rooms	Part of the rooms are equipped or easily adaptable (dedicated furniture, space dimension) to host patients with special needs such as pregnant women, paediatric patients, a parent with child.	Few rooms are equipped or easily adaptable (dedicated furniture, space dimension) to host patients with special needs such as pregnant women, paediatric patients, a parent with child.	NO rooms are equipped or easily adaptable (dedicated furniture, space dimension) to host patients with special needs such as pregnant women, paediatric patients, a parent with child.					
4.3.7	Toilet facility (further criteria in section 2.4)	Both suspect and confirmed cases have private toilet OR Suspect cases rooms have private toilets and shared toilets are available for confirmed cases rooms.	Confirmed cases room have shared toilets that fully meet the criteria listed in section 2.4.5 OR not all the suspect cases room have individual toilets.	Suspect and confirmed cases share toilets.					
4.3.8	Floor surface	Single bedroom is 12 m ² / bay dimensions is 3.45 x 3.6 metres, objects around the bed do not represent an obstacle for the staff to access the patient.	Single bedroom is between 9 and 12 m ² / bay dimensions is less than 3.45 x 3.6 metres but more than 2.5 x 3 metres and objects around the bed that may represent an obstacle for the staff to access the patient.	Single bedroom is less than 9 m ² / bay dimensions is less than 2.5 x 3 metres per patient.					
4.3.9	Donning/ doffing (putting on/ taking off PPE) location	There are dedicated donning and doffing spaces for the ward that are strategically located between the low-risk zone and the high-risk zone.	There are NO designated spaces for donning and doffing dedicated to the ward, but staff can use donning and doffing spaces, that are strategically located between the low-risk zone and the high-risk zone, and that are in proximity to the ward (less than 30 metres).	There are NO designated spaces for donning and doffing dedicated to the ward and staff put on and take off the PPE in donning and doffing rooms far from the ward (more than 30 metres) OR donning and doffing are not strategically located between the low-risk zone and the high-risk zone.					

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
4.3.10	Donning dimensions and furniture	The donning area is big enough to ensure space for shelves for PPE storage, bins, mirror, bench/chair, small table. The dimension of the donning and doffing areas also ensures optimal free movement of 3 staff (at least 20 m²).	The donning area is big enough to ensure space for shelves for PPE storage, bins, mirror, a chair. The dimension of the donning and doffing areas also ensures free movement of 3 staff (between 16 m² and 20 m²).	The donning is not big enough to ensure space for basic furniture and PPE storage AND staff has no sufficient space to properly perform donning procedure (less than 16 m²).					
4.3.11	Doffing dimensions and furniture	The doffing area is big enough to ensure space for full rotation with open arms for 1 staff, 2 metres distance with the second staff monitoring, bins and handwashing station The doffing area (1 line) is at least 12 m².	The doffing area is big enough to ensure 2 metre distance with the second staff monitoring, bins and handwashing station. The full rotation with open arms of staff taking off the PPE is difficult The doffing area (1 line) is between 10 m² and 12 m².	The doffing area is not big enough to ensure full movement of the staff OR to ensure 2 metres distance between the 2 staff OR to host bins and handwashing station. The doffing area (1 line) is less than 10 m².					
4.3.12	Doffing lines	There are at least 2 doffing lines plus 1 line every 10 beds.	There are at least 2 doffing lines plus 1 line every 15 beds.	There is only 1 doffing line.					
4.3.13	Surge plan	Extra rooms are available in case the beds demand increase; it is equipped and ready at any time.	Extra rooms are available in case the beds demand increase but it is NOT equipped.	There is NO extra rooms are available in case the beds demand increase.					
4.3.14	Discharge location	A discharge shower is available for every ward, and every negative tested patient or survivor is passing through it before exit the treatment ward. Its location enhances the unidirectional flow, and it is located near the treatment ward (less than 10 metres).	A discharge shower is available, and every negative tested patient or survivor is passing through it before exit the facility. Its location enhances the unidirectional flow, but it is located far (more than 10 metres) from the treatment ward.	A discharge shower is not available OR it is available but not every negative tested patient or survivor pass through it before exit the facility.					
4.3.15	Discharge dimensions and furniture	The discharge shower is at least 4 m ² . There is space for dressing, undressing, a shower and to ensure the flow is unidirectional.	The discharge shower is between 3 m ² and 4 m ² . There is space for dressing/undressing, and a shower.	The discharge shower is less than 3 m ² . Shower and changing zone are sharing the same area.					

4.4 Morgue

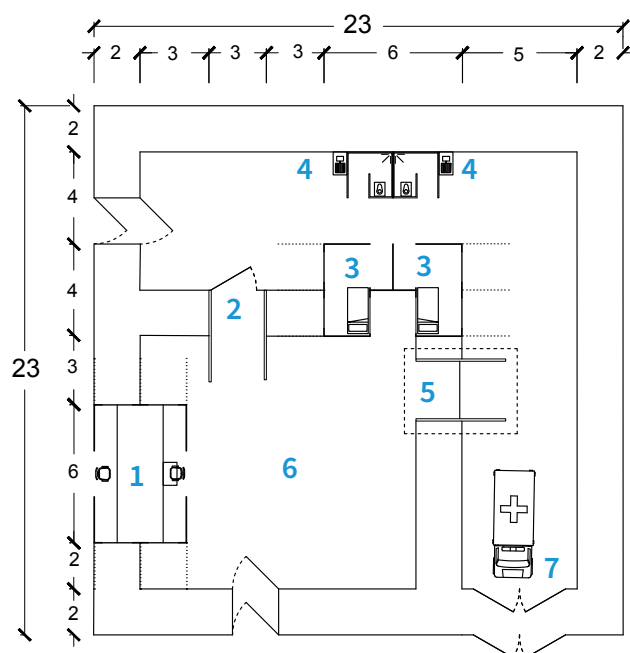
No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
4.4.1	Location and access	The morgue is easily accessible from all the wards but not visible If the onsite morgue or mortuary allows visitors: Visitors can access the morgue directly from outside the centre and see the body from a distance of 1 metre, or through a transparent window; the morgue has a dedicated exit.	The morgue is easily accessible from all the wards but visible If the onsite morgue or mortuary allows visitors: Visitors CANNOT access the morgue directly from outside the centre to see the body; the morgue has a dedicated exit.	The morgue is NOT easily accessible from all the wards If the onsite morgue or mortuary allows visitors: Visitors CANNOT access the morgue directly from outside the centre to see the body; the morgue does NOT have a dedicated exit.					
4.4.2	Area distribution <i>(only if the onsite morgue or mortuary allows visitors)</i>	There is a dedicated area for temporary storage of bodies, an area to show the body to family members (without visibility of the storage area).	There is a dedicated area for temporary storage of bodies, an area for the family with no good visibility or without privacy.	There is dedicated area for temporary storage of bodies but no family area.					
4.4.3	Distance/ barrier visitors	There is a physical barrier assuring 2 metres distance OR a transparent surface to allows visitors to safely see the body.	There is NO physical barrier assuring 2 metres distance, but staff accompany relatives during the visit to the morgue.	Relatives CANNOT see the body.					

4.5 Other services

No.	Area	Meets target +++ (3)	Partially meets target ++ (2)	Does not meet target + (1)	Meets	Partial	Does not	DK	NA
4.5.1	Laundry	The facility is equipped with a dedicated functioning laundry service that is properly sized for the centre capacity. It includes cleaning and disinfection area, drying area, folding & storage area. It is on-site, in low-risk area, with running water and chlorine supply.	The facility is equipped with a dedicated functioning laundry service that is properly sized for the centre capacity. One or more of these functional areas is missing or inadequate: • Cleaning and disinfection area is inadequate or lacks proper equipment, OR • Drying area is insufficient, OR • Folding & storage area is lacking or inadequate for clean PPE inventory.	The facility has no on-site laundry OR missing critical infrastructure: • No running water access in laundry area, OR • No chlorine OR disinfectant supply available OR a laundry service exists but is NOT operational or functioning.					
4.5.2	Storage/warehouse	There is a logistic storage for consumables. It is easily accessible both from outside and inside the treatment centres.	There is a logistic storage for consumables. But it is only accessible from inside the treatment centres, forcing the supplier to access the treatment centres compound to deliver the items.	There is no logistic storage for consumables.					
4.5.3	Pharmacy	There is a pharmacy available. It is strategically located near the treatment area, and the indoor temperature is compliant with the stored drugs requirements.	There is a pharmacy available, but it is NOT strategically located near the treatment area OR the indoor temperature is not compliant with the stored drugs requirements.	There is no pharmacy available, drugs are randomly stored in other treatment centres areas.					

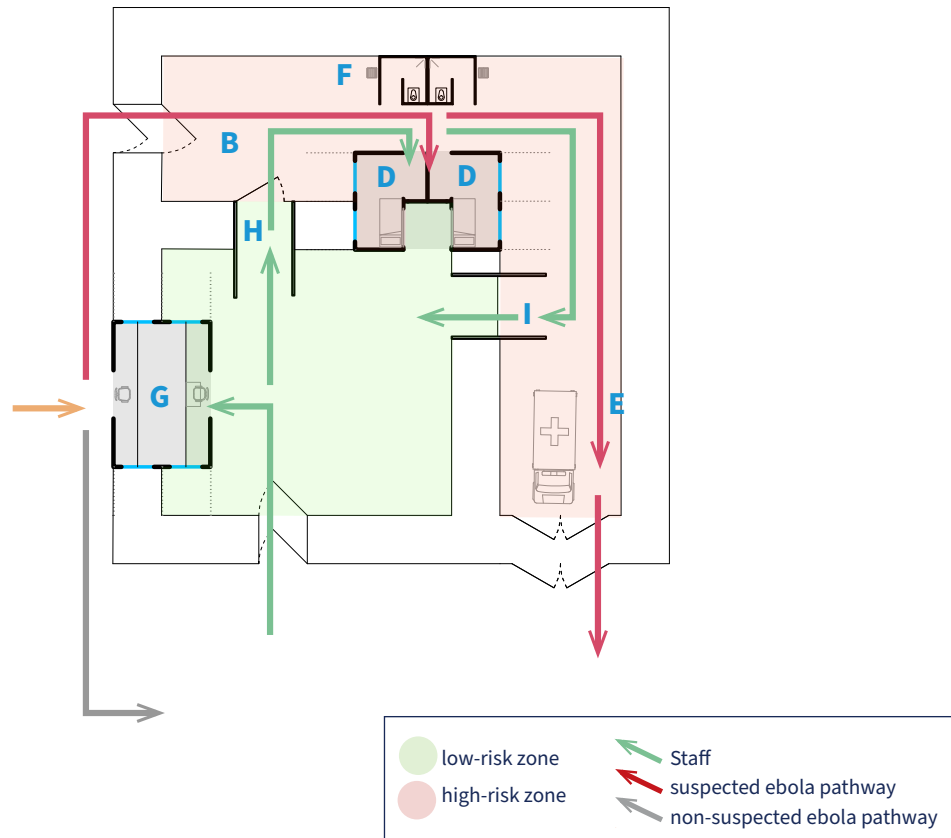
Annex 4. Layout examples

Screening and temporary isolation with tents



- | | |
|---|------------------|
| ①. screening point | ⑤. doffing |
| ②. donning | ⑥. staff area |
| ③. temporary isolation room (patient's room) | ⑦. ambulance bay |
| ④. patients dedicated toilets and handwashing | |

The dimensions indicated in the drawing are in metres.

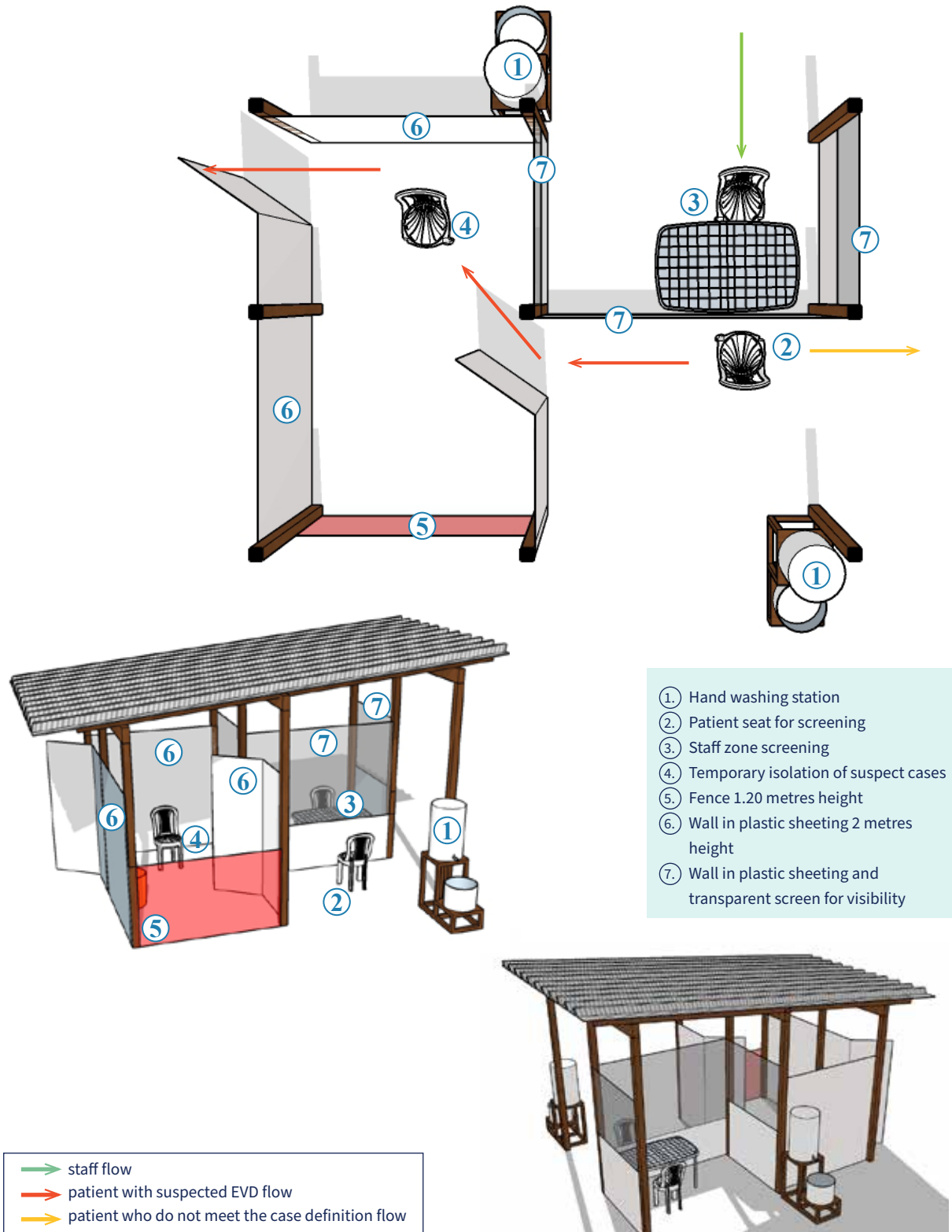


Any person accessing the health facility undergo the screening (A). If the Ebola case definition is met, the patient accesses the temporary isolation area (B). A hand hygiene station is available outside the screening station. If the case definition is not met, the patient accesses the health facility (C). Clinical assessment can take place inside the patient's room (D). There, the patient waits referral to the appropriate treatment facility (E). Individual toilets and handwashing stations are available (F). The screening station (G) is equipped with a double fence to ensure 2 metres distance between the patient and the staff.

Before accessing the patient's area, the staff put on the PPE in the donning area (H). Before returning to the staff area, the staff take off the PPE in the doffing area (I). Staff toilets and services are available in the main health facility.

SITE DIMENSIONS 23 x 23 metres

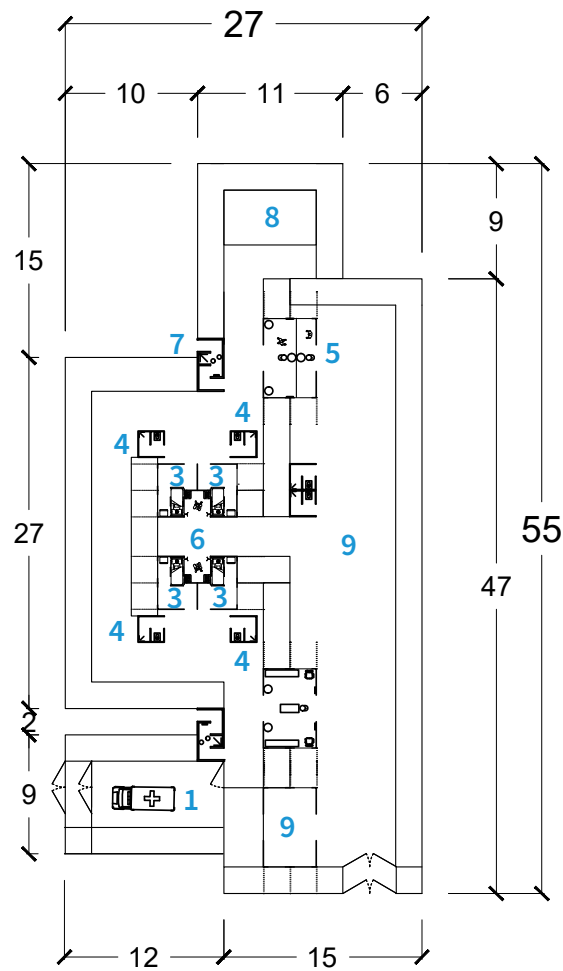
Screening and temporary isolation with local materials



The basic screening is a simple solution optimal for the installation at the entrance of any small or medium health facility. A transparent screen characterizes the screening station, to ensure no contact occurs between patients and staff. Patients who meet the case definition move to the temporary isolation area. This area is closed with 2 metres high partitions.

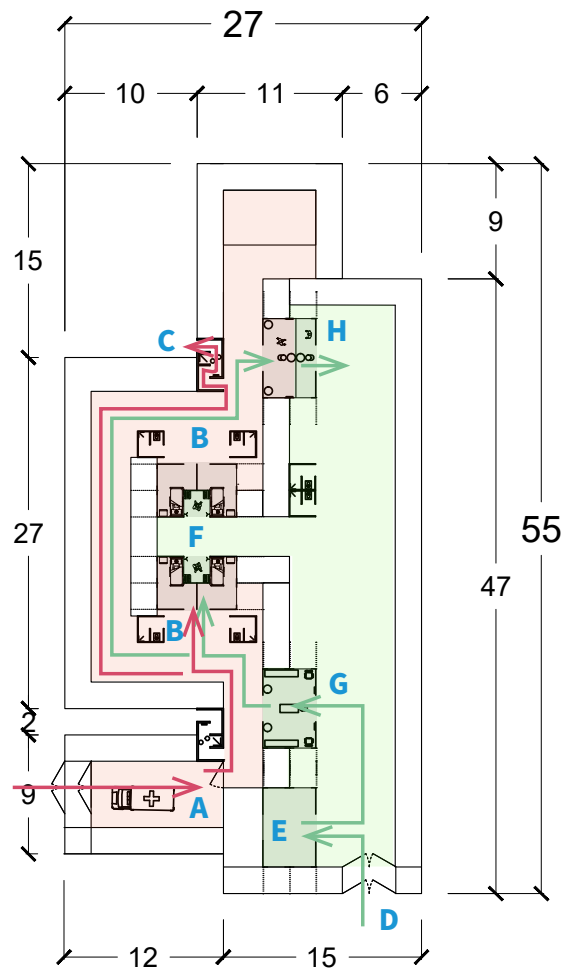
SITE DIMENSIONS: 5 x 4 metres

Ebola or Marburg treatment centre with 4 beds



- | | |
|---|-----------------------------------|
| ① ambulance bay | ⑥ central corridor for visibility |
| ② donning | ⑦ discharge shower |
| ③ patient room | ⑧ waste zone |
| ④ patients dedicated toilets, showers and handwashing | ⑨ staff zone |
| ⑤ doffing | |

The dimensions indicated in the drawing are in metres.

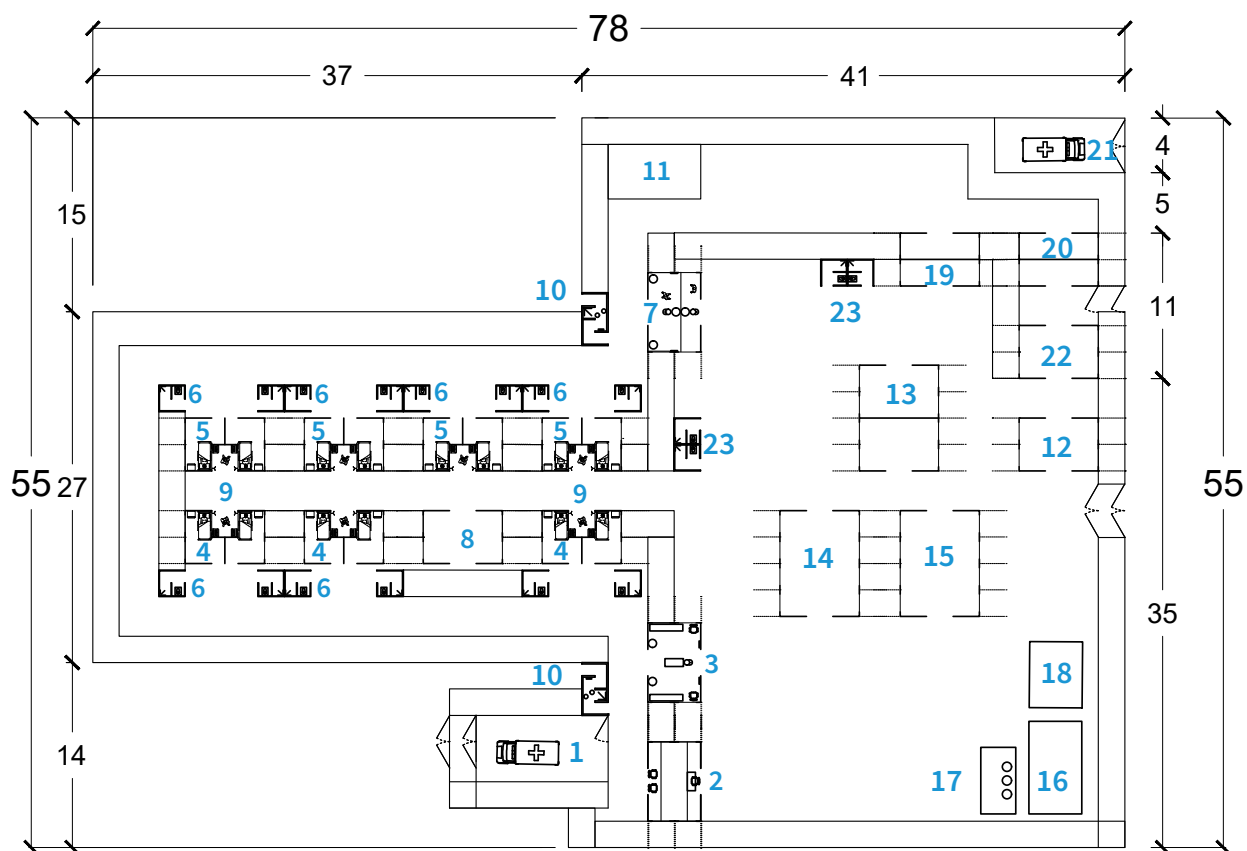


The dimensions indicated in the drawing are in metres.

Patients, either arriving autonomously or by ambulance, access the centre through the dedicated patient entrance (A). Patients are admitted to the treatment ward, where they are accommodated in individual rooms with dedicated toilets (B). This minimum configuration can accommodate only suspects, only confirmed or both suspect and confirmed cases, according to the needs, thanks to the individual rooms with private toilets. Recovered patients leave the centre through the discharge shower (C). Staff access the treatment unit through a dedicated entrance (D). They change into scrubs in the changing room (E) before proceeding according to their assigned duties. Medical staff can monitor patients' conditions through transparent observation screens (F) (patient liner). Before entering the patient care area, staff put on the PPE in the donning area (G). To exit the high-risk zone, staff take off their PPE in the doffing area (H) before returning to the low-risk area.

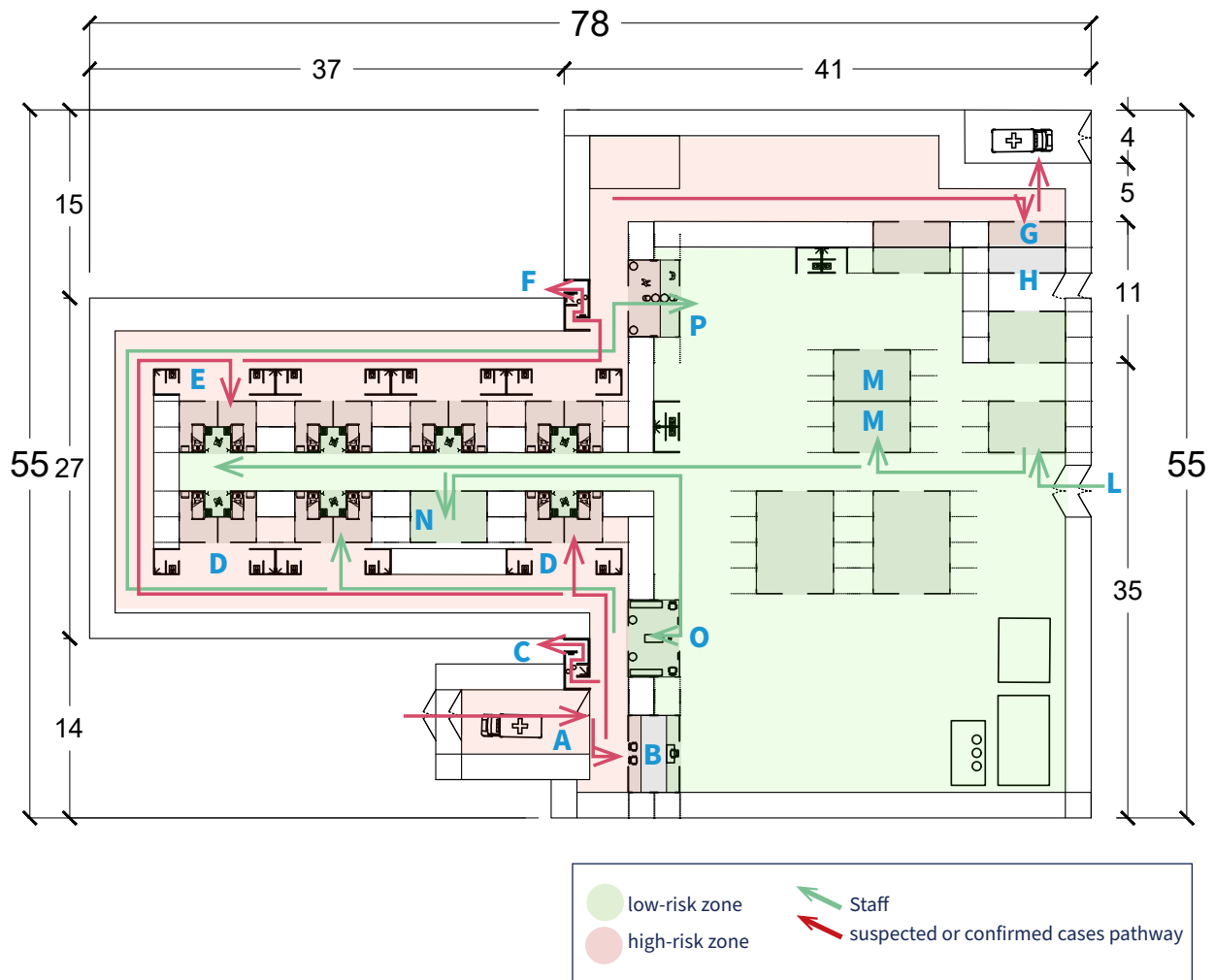
SITE DIMENSIONS 27 x 55 metres

Ebola or Marburg treatment centre with 14 beds



- | | | |
|---|--|---------------------------|
| ① ambulance bay | ⑩ discharge shower | ⑳ morgue |
| ② screening | ⑪ waste zone | ㉑ car bay |
| ③ donning | ⑫ staff reception and scrub distribution | ㉒ psychosocial support |
| ④ patient room suspects | ⑬ staff changing room | ㉓ staff toilet and shower |
| ⑤ patients room confirmed | ⑭ laundry and drying | |
| ⑥ patients dedicated toilets, showers and handwashing | ⑮ warehouse and pharmacy | |
| ⑦ doffing | ⑯ water bladder | |
| ⑧ nursing station | ⑰ chlorination | |
| ⑨ central corridor for visibility | ⑱ generator | |
| | ⑲ laboratory | |

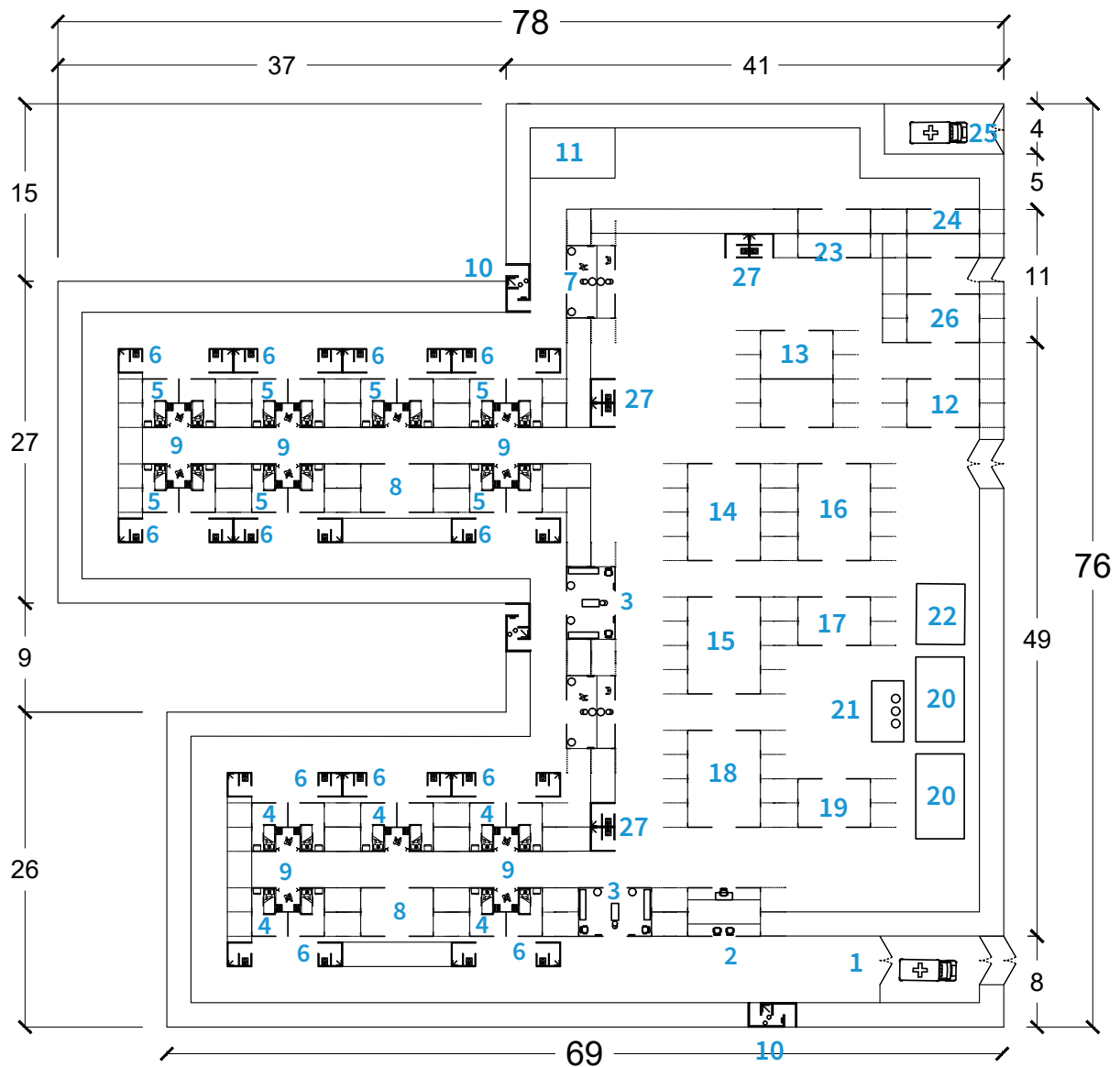
The dimensions indicated in the drawing are in metres.



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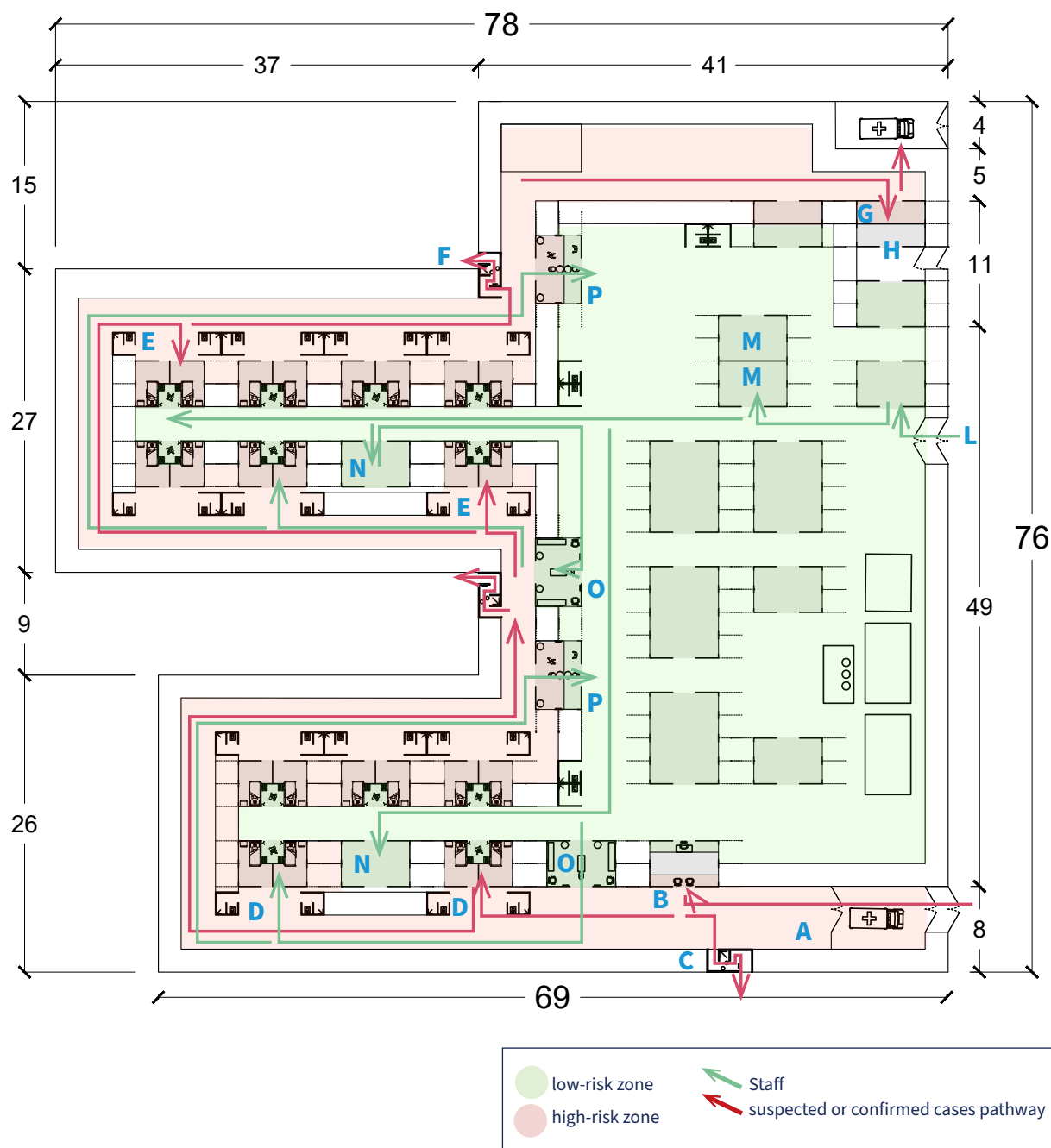
Patients, either arriving autonomously or by ambulance, access the centre through the dedicated patient entrance (A) to undergo screening (B). Patients who do not meet the case definition leave the centre through the dedicated discharge shower (C). Patients who meet the case definition are admitted to the suspected cases zone, where they are accommodated in individual rooms with dedicated toilets (D). Confirmed cases are then transferred to the confirmed cases rooms (E). Recovered patients leave the centre through the discharge shower (F), while deceased patients are transferred to the morgue (G). Family members may view the deceased from outside the facility through a transparent screen (H), after which the body is handed over through the designated transfer point in the fence (I) for transportation and safe burial. Staff access the treatment unit through a dedicated entrance (L). They change into scrubs in the changing room (M) before proceeding according to their assigned duties. Medical staff report to the nursing station (N), from where patient conditions can be monitored through transparent observation screens. Before entering the patient care area, staff put on the PPE in the donning area (O). To exit the high-risk zone, staff take off their PPE in the doffing area (P) before returning to the low-risk area.

Ebola or Marburg treatment centre with 24 beds



- | | | |
|---|--|---------------------------|
| ① ambulance bay | ⑩ discharge shower | ⑳ water bladder |
| ② screening | ⑪ waste zone | ㉑ chlorination |
| ③ donning | ⑫ staff reception and scrub distribution | ㉒ generator |
| ④ patient room suspects | ⑬ staff changing room | ㉓ laboratory |
| ⑤ patients room confirmed | ⑭ laundry | ㉔ morgue |
| ⑥ patients dedicated toilets, showers and handwashing | ⑮ drying | ㉕ car bay |
| ⑦ doffing | ⑯ warehouse | ㉖ psychosocial support |
| ⑧ nursing station | ⑰ pharmacy | ㉗ staff toilet and shower |
| ⑨ central corridor for visibility | ⑱ staff meeting room | |
| | ⑲ WASH-log office | |

The dimensions indicated in the drawing are in metres.



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